20 Turbot in the North Sea

tur.27.4 – *Scophthalmus maximus* in Subarea 4

This report presents the stock assessment carried out for turbot (*Scophthalmus maximus*) in Subarea 4 in 2024. Turbot in 27.4 was upgraded to a Category 1 stock including reference points and a short-term forecast during the 2018 inter-benchmark (ICES, 2018b). The assessment is conducted and advice for turbot in 27.4 is provided based on the assessment configuration and short-term forecast derived during the 2025 benchmark. Reference points were recalculated during WGNSSK 2025 as a result of an error found in the 2023 catch raising procedures.

20.1 General

20.1.1 Biology and ecosystem aspects

Turbot is broadly distributed from Iceland in the North, along the European coastline, to the Mediterranean and Adriatic Sea in the south. In general, turbot is a rather sedentary species, but there are some indications of migratory patterns. For example, in the North Sea, migrations from the nursery grounds in the south-eastern part to more northerly areas have been recorded. IBPNEW (ICES, 2012) concluded that turbot in the North Sea (Subarea 4) can be considered as a distinct stock for management purposes. However, recent genetic studies and species distribution mapping show that the Skagerrak part of the stock could potentially be merged with the North Sea stock and the Kattegat with the Baltic Sea stock (ICES, 2020).

Turbot is typically found at a depth range of 10 to 70 m, on sandy, rocky or mixed bottoms and is one of the few marine fish species that inhabits brackish waters. It is a typical visual feeder and could be regarded as a top predator. Turbot feeds mainly on bottom living fishes (e.g. common gadoids, sandeel, gobies, sole, dab, dragonets, sea breams, etc.) and small pelagic fish (e.g. herring, sprat, boarfish, sardine) but also, to a lesser extent, on larger crustaceans and bivalves.

20.1.2 Fisheries

In the 1950s, the UK was the biggest contributor to the landings (~50% of the landings). In recent years, most of the landings stem from the Netherlands (~50%). Turbot is mostly caught in trawls of mixed fisheries, such as the 80 mm beam trawl fleet (BT2) fishing for sole and plaice (representing 64% of Dutch turbot landings in 2024). In Denmark, the second largest contributor to the landings in recent times, there is a directed fishery for turbot using gillnets (13% of the total landings in 2024).

See the Stock Annex for more details.

20.1.3 Management

The turbot stock was previously managed using a combined TAC for turbot and brill, set in Subarea 4 and in UK waters of Division 2.a. This management area (particularly the inclusion of a part of Division 2.a) did not correspond to either of the stock areas defined by ICES for turbot and brill.

As of 2024, the combined turbot and brill TAC technically remains in place for Subareas 2.a (UK only) and 4 (EU + UK). However, the TAC is now divided into separate allocations for the UK and EU member states for both species, with the TAC for turbot set at 2038 tonnes in 2025 (EU–UK Joint Committee, 2023). This updated approach effectively abolishes the combined TAC in practical terms.

As a primarily bycatch species, regulations relating to effort restrictions for the primary métiers catching turbot (e.g. beam trawlers) are likely to impact the stock. Fishing effort has been restricted in the past for demersal fleets in a number of EC regulations (e.g. EC Council Regulation Nos. 2056/2001, 51/2006, 41/2007, and 40/2008).

The Dutch Producer Organisations (POs) have introduced a minimum landings size of 27 cm in 2013. In 2016, catches of turbot increased substantially and the minimum landing size was increased to 30 cm at first, followed by a further increase to 32 cm in May 2016. In the summer of 2016, the POs decided to prohibit landing the smallest market category and in October and November the weekly landings were capped to respectively 375 kg and 600 kg wk^{-1}. These measures were taken to keep the landings in line with the available national quota. In 2018, the TAC for turbot and brill was substantially increased; however, Dutch PO measures were still in place with a minimum landing size of 30 cm and limiting the landings to 2000 kg wk^{-1}. Since 2019, the PO measures were relaxed due to the sufficiently available quota. In 2025, the minimum landing size was lowered to 25 cm.

Measures taken by the Dutch Producer Organisations from 2016 up to present.

Dutch PO-measures			
Year	Date	Max kg per week/trip	MLS
2016	January - March	-	27 cm
2016	April – May	-	30 cm
2016	May – September	-	32 cm
2016	October – November	375 kg	32 cm
2016	November – December	600 kg	32 cm
2017	January – February	-	32 cm
2017	March – October	800 kg	32 cm
2017	November - December	2000 kg	30 cm
2018	January – August	2000 kg	30 cm
2018	September - October	2500 kg	30 cm
2018	October - December	3000 kg	27 cm
2019	January – December	3000 kg	27 cm
2020	January - December	3000 kg	27 cm
2021	January – December	3000 kg	27 cm
2022	January – December	3000 kg	27 cm

Dutch PO-measures			
2023	January – December	3000 kg	27 cm
2024	January – December	3000 kg	27 cm
2025	January - March	3000 kg	27 cm
2025	March -	3000 kg	25 cm

20.2 Data used in the assessment

Following the benchmark conducted in early 2025 (ICES, 2025), the assessment of North Sea turbot requires three main types of data:

Catch data: estimates of removals of turbot by the fishery.

Survey data: indices of trends in population abundance over time from fisheries independent sources, respectively.

Biological data: estimates and/or assumptions on growth, maturation and natural mortality.

Since the assessment is age-based, data for the above is required for each age. See the Stock Annex for more details on the data used in the assessment, sources and historical values.

20.2.1 Catch data

Figure 20.1 and Table 20.1 show the trend in total landings (InterCatch) and discards (InterCatch) over time. ICES estimated landings of turbot decreased during the 1990s and 2000s. From 2012 – 2020 landings fluctuated around 3000 tonnes but have declined from 2659 t in 2021 to 1662 t in 2023. In this period, effort by the Dutch beam trawl fleet, which contributes most to the landings (ca. 45%), has decreased notably due to restricting quota, rising energy prices and a resulting large-scale decommissioning of the fleet. In 2024, landings were 1750 t.

In 2025, UK England updated its 2021 to 2023 data. This change as well as updated raising procedures resulted in an increase of 12 tonnes for total discards <1 t in total landings in 2021, an 11 t increase in landings and <1 t increase in discards in 2022 and a 6 t and 8 t increase in landings and discards respectively in 2023. An error in the InterCatch age-allocation raising procedure was found in the 2023 data year. After fixing the error and re-calculating the age allocation specified in the landings, the age distribution in the landings shifted from younger to older ages in 2023 (Figure 20.3).

Since 2018, official landings in Subarea 4 decreased gradually from 3228 tonnes to 1647 tonnes in 2023 (Table 20.2). Official landings of turbot in Subarea 4 were 1754 t in 2024 (Figure 20.4). Since 2016, the combined TAC for turbot and brill has not been fully utilized. In 2024, ~57% of the combined TAC (3606 tonnes) was taken, of which turbot had the largest share (73%). In 2024, 86% of the TAC for turbot was taken.

Landings in numbers-at-age are presented in Table 20.3 and Figure 20.2. Following a decrease in minimum market size for turbot in the Netherlands in 2002, there has been a notable increase in the amount of age 1 and 2 turbot landed, accounting for half of the landings in some years. This proportion has been decreasing in recent years due to some poor recruitment in 2012, 2013, 2016, and 2021. Since turbot are only fully mature-at-age 4, a high proportion of immature fish are part of the landings. In 2021, there were no landings of age 1 suggesting a weak year class. The landings of age 1 fish in 2022-2024 were similar to those of the 2018-2020-year classes, suggesting stronger year classes coming through in the landings.

The weights-at-age in the landings of turbot in Subarea 27.4 (Table 20.4) come from the "weca" file of the InterCatch landings export. These are measured weights from the various national catch and market sampling programmes. Mean stock weights-at-age (Table 20.5) are the average weights from the 2nd quarter landings and are derived from the "Catch and Sample Data Table" file from InterCatch. Given the lack of weight data in the period 1991–2003, modelling was required to infer the trend in landings and stock weight-at-age data (Table 20.6 and 20.7, respectively; more details in the stock annex).

20.2.2 Discards

The assessment of this stock does not include discards as there is very limited age sampling of the discards and discard rates are generally limited to under 10%. In 2018, 4% of the imported discard data were sampled, coming from discards of Danish (< 8 fish per métier) and Belgian beam trawl fleets (138 fish). These data were considered insufficient to be used in the age allocation of international discards. In 2018-2023, no age structure information was submitted for the discard estimates as sample sizes were too low for raising and to be submitted to InterCatch. In 2024, 3% of the discards was age-sampled, from Belgian 80 mm beam trawlers. As discards are not included in this assessment, discard weight-at-age from these limited samples are not presented.

There was a sudden increase in the landings of age two turbot following the decrease in minimum market size in the Netherlands in 2002. Given that there was no known change in the fishing behaviour of the main fleets at this time, this could indicate that, previously, more age 1 fish must have been caught than were actually landed. These were either discarded or, as a much-sought-after fish, kept by the fishermen for personal use. This would mean that the discards could be underestimated in the period up to 2002 relative to the period following this. Alternatively, subsequent to the change in MLS, more targeting of small turbot may have occurred. Without a useable time-series of discards before and after this change it is difficult to determine which of these explanations holds.

The discard rate (discards: 81 t / (discards + landings: 1831 t)) was 4.4% in 2024 marking a decrease in turbot discards compared to 2023 (10.5%).

In 2024, BMS landings were reported by the UK (England/Wales). However, the submitted values were negligible (12 kg) and were not raised in InterCatch.

20.2.3 Logbook registered discards

In 2024, no logbook registered discards were reported to InterCatch. They are not raised.

20.2.4 InterCatch raising and allocation procedures

InterCatch was used for the first time for the North Sea turbot stock at WGNSSK 2014, and has been used since.

In 2024, discard estimates were submitted to InterCatch by six countries, with three countries reporting non-zero values. Where possible, discards were raised within métier by quarter. Unsampled discards of beam trawlers with a mesh size < 100 mm were raised using sampled catch of beam trawlers with a mesh size < 100 mm within quarter. Unsampled discards of beam trawlers with a mesh size > 100 mm were raised using sampled discards of beam trawlers with a mesh size < 100 mm for all quarters given the absence of samples in the > 100 beam trawl fleet. For all other gears (mainly towed gears such as OTB and SSC), discards were raised using discard rates from all other gears for all quarters.

Unsampled fleets*	Sampled fleets**
TBB < 100mm	Within metier, by quarter
TBB > 100mm	From TBB < 100mm, all quarters
Rest (GNS, OTB, etc.)	All (Mainly OTB and SSC), all quarters

*** Unsampled fleets are those fleets for which no discards are submitted.**

**** Sampled fleets are those fleets for which discards ratios are known.**

Out of the 81 tonnes of estimated discards, 49 tonnes (61%) were reported data and 32 tonnes (39%) were raised in InterCatch. The proportion of landings with discards associated (same strata) was 52%. The age sampling of the landings in 2024 covered 37% of all landings.

For the landings, Dutch (for data from 2004–present), Danish (2014–present) and Belgian (2017–present) samples, accounting for auctions, quarters and market categories, are provided. The number of age samples gradually increased from 2018 (2359 samples), to 6068 samples in 2024. This was however a decrease since 2023 (11027). In total, Denmark supplied 5511 samples collected in various metiers in 2024, while the Dutch (421) samples only consist of the TBB_DEF_70-99 fleet. All data are used for estimating the age structure of the landings. Prior to 2004, the landings-at-age information is from an old Dutch monitoring scheme from the 1980s. Figure 20.5 shows the métiers with age distributions for the landings in 2024. Allocations to calculate the age structure were done for discards and landings separately and were done within métier per quarter where possible. If by quarter was not possible, available quarters were grouped. Very limited age structure information for discards was available in 2024, and was only obtained from Belgian samples. Given the limited extent of this sampling, the available age samples of the landings were used for the discards, although not used in the assessment or forecast.

Unsampled fleets*	Sampled fleets**
TBB < 100mm	Within métier, all quarters
TBB > 100 mm	Within métier, all quarters
Others	All métiers, all quarters

*** Unsampled fleets are those fleets for which no age structure is known.**

**** Sampled fleets are those fleets for which age structure is known.**

20.2.5 Biological data

All biological data used in the assessment are presented in Tables 20.5, 20.7, 20.8 and 20.9.

Weight-at-age

Weights from landings are used for catch weights-at-age, and Q2 landings are used for stock weights-at-age. Available weight at age data is very noisy, due to low sample sizes for most ages (Figure 20.6). The data that are available, and trends in other flatfish species in the same areas, suggest that there have been potentially significant changes in weight-at-age over time. At the

2025 benchmark, a method was accepted to estimate catch and stock weights-at-age internally in SAM using Gaussian Markov Random Field method accounting for trends in years and cohorts. The results indicate an increase in weight-at-age from the start of the time series, peaking in the early 1990s. Following this period, weights-at-age showed a declining trend until 2019 but have been gradually increasing since then (Figure 20.6).

Maturity

In this assessment individuals are assumed to be 100% mature at age 5, 98% at age 4, 58% at age 3, 1% at age 2 and 0% for age 1. See Stock Annex for full details.

Natural mortality

An age varying natural mortality vector was implemented during the 2025 benchmark, with M decreasing from 0.61 at age 1 to 0.27 at age 8+. See Stock Annex for full details.

20.2.6 Survey data

For the 2025 turbot benchmark, new modelled indices for North Sea turbot were developed using the R package surveyIndex. A new combined index was developed to replace the previously used stratified mean CPUE-based index from the Dutch offshore BTS survey (BTS-ISIS) in the North Sea turbot assessment. This new index incorporates BTS data from all relevant North Sea countries (NL, GB, BE, and DE). During the 2025 benchmark, it was decided to combine the new BTS modeled index with information derived from the sole net survey (SNS) and demersal young fish survey (DYFS) effectively combining all relevant North Sea beam trawl surveys carried out by scientific vessels into one comprehensive index titled "BTS+COAST". The new index also replaced the previous CPUE-based index derived from the SNS survey. The index runs from 1991 to 2024 (Figure 20.7, Table 20.10). A model-based index for BSAS for the years 2019-2024 was developed using the same methodology as for the other scientific surveys (Figure 20.7, Table 20.11). See Stock Annex for full details.

The BTS+COAST survey index shows good internal consistency for ages 1-5 but poor tracking of ages 6-7 (Figure 20.8). The BSAS survey index seems to perform well for older ages, although there are issues in the cohort tracking of ages 3-4 (Figure 20.8). The consistency of the index will be monitored as more years of data are added.

20.3 Stock assessment model

Since the inter-benchmark protocol of 2017, the SAM assessment model (SAM, FLSAM; Nielsen and Berg, 2014; Berg and Nielsen, 2016) is used with an observation correlation structure. The model settings were re-evaluated during the 2025 benchmark (ICES, 2025). More details can be found in the Stock Annex.

20.3.1 Model settings

The assessment model was conducted using the settings and configuration given below. Details of the assessment model can be found in the Stock Annex and benchmark report (ICES, 2025).

Model settings used in the final assessment

SAM configuration file

```
$minAge
# The minimium age class in the assessment
 1
$maxAge
# The maximum age class in the assessment
 8
$maxAgePlusGroup
# Is last age group considered a plus group for each fleet (1 yes, or 0 no).
 1 1 1
$keyLogFsta
# Coupling of the fishing mortality states processes for each age (normally only
# the first row (= fleet) is used).
# Sequential numbers indicate that the fishing mortality is estimated individually
# for those ages; if the same number is used for two or more ages, F is bound for
# those ages (assumed to be the same). Binding fully selected ages will result in a
# flat selection pattern for those ages.
  0  1  2  3  4  5  6  6
 -1 -1 -1 -1 -1 -1 -1 -1
 -1 -1 -1 -1 -1 -1 -1 -1
$corFlag
# Correlation of fishing mortality across ages (0 independent, 1 compound symmetry,
# 2 AR(1), 3 separable AR(1).
# 2: AR(1) first order autoregressive - similar ages are more highly correlated than
# ages that are further apart, so similar ages have similar F patterns over time.
# if the estimated correlation is high, then the F pattern over time for each age
# varies in a similar way. E.g if almost one, then they are parallel (like a
# separable model) and if almost zero then they are independent.
2
$keyLogFpar
# Coupling of the survey catchability parameters (normally first row is
# not used, as that is covered by fishing mortality).
 -1 -1 -1 -1 -1 -1 -1 -1
  0  1  2  3  4  5  5 -1
  6  7  8  9 10 11 12 12
$keyVarF
# Coupling of process variance parameters for log(F)-process (Fishing mortality
# normally applies to the first (fishing) fleet; therefore only first row is used)
  0  1  2  2  3  3  3  3
 -1 -1 -1 -1 -1 -1 -1 -1
 -1 -1 -1 -1 -1 -1 -1 -1
$keyVarLogN
# Coupling of the recruitment and survival process variance parameters for the
# log(N)-process at the different ages. It is advisable to have at least the first age
# class (recruitment) separate, because recruitment is a different process than
# survival.
 0 1 1 1 1 1 1 1
$keyVarObs
# Coupling of the variance parameters for the observations.
# First row refers to the coupling of the variance parameters for the catch data
# Second and further rows refers to coupling of the variance parameters for the indices
  0  1  1  1  1  1  2  2
```

3 4 4 4 4 5 5 -1
6 7 7 7 7 7 7 7
$obsCorStruct
Covariance structure for each fleet ("ID" independent, "AR" AR(1), or "US" for unstructured). | Possible values are: "ID" "AR" "US"
"ID" "AR" "ID"
$keyCorObs
Coupling of correlation parameters can only be specified if the AR(1) structure is chosen above.
NA's indicate where correlation parameters can be specified (-1 where they cannot).
#1-2 2-3 3-4 4-5 5-6 6-7 7-8
NA NA NA NA NA NA NA
0 0 0 0 0 0 0
NA NA NA NA NA NA NA
$stockRecruitmentModelCode
Stock recruitment code (0 for plain random walk).
0
$fbarRange
lowest and higest age included in Fbar
2 6
$fixVarToWeight
If weight attribute is supplied for observations this option sets the treatment (0 relative weight, 1 fix variance to weight). Can be specified fleetwise.
0 0 0
$fracMixF
The fraction of t(3) distribution used in logF increment distribution
0
$fracMixN
The fraction of t(3) distribution used in logN increment distribution (for each age group)
0 0 0 0 0 0 0 0
$fracMixObs
A vector with same length as number of fleets, where each element is the fraction of t(3) distribution used in the distribution of that fleet
0 0 0
$stockWeightModel
Integer code describing the treatment of stock weights in the model (0 use as known, 1 use as observations to inform stock weight process (GMRF with cohort and within year correlations)), 2 to add extra correlation to plusgroup
1
$keyStockWeightMean
Coupling of stock-weight process mean parameters (not used if stockWeightModel==0)
0 1 2 3 4 5 6 7
$keyStockWeightObsVar
Coupling of stock-weight observation variance parameters (not used if stockWeightModel==0)
0 0 0 0 0 0 0 0
$catchWeightModel
Integer code describing the treatment of catch weights in the model (0 use as known, 1 use as observations to inform catch weight process (GMRF with cohort and within year correlations)), 2 to add extra correlation to plusgroup
1
$keyCatchWeightMean
Coupling of catch-weight process mean parameters (not used if catchWeightModel==0)
0 1 2 3 4 5 6 7
$keyCatchWeightObsVar

```
# Coupling of catch-weight observation variance parameters (not used if catchWeightModel==0)
  0  0  0  0  0  0  0  0
```

SAM configuration file

```
$minAge
# The minimium age class in the assessment
 1
$maxAge
# The maximum age class in the assessment
 8
$maxAgePlusGroup
# Is last age group considered a plus group for each fleet (1 yes, or 0 no).
 1 1 1
$keyLogFsta
# Coupling of the fishing mortality states processes for each age (normally only
# the first row (= fleet) is used).
# Sequential numbers indicate that the fishing mortality is estimated individually
# for those ages; if the same number is used for two or more ages, F is bound for
# those ages (assumed to be the same). Binding fully selected ages will result in a
# flat selection pattern for those ages.
  0  1  2  3  4  5  6  6
 -1 -1 -1 -1 -1 -1 -1 -1
 -1 -1 -1 -1 -1 -1 -1 -1
$corFlag
# Correlation of fishing mortality across ages (0 independent, 1 compound symmetry,
# 2 AR(1), 3 separable AR(1).
# 2: AR(1) first order autoregressive - similar ages are more highly correlated than
# ages that are further apart, so similar ages have similar F patterns over time.
# if the estimated correlation is high, then the F pattern over time for each age
# varies in a similar way. E.g if almost one, then they are parallel (like a
# separable model) and if almost zero then they are independent.
2
$keyLogFpar
# Coupling of the survey catchability parameters (normally first row is
# not used, as that is covered by fishing mortality).
 -1 -1 -1 -1 -1 -1 -1 -1
  0  1  2  3  4  5  5 -1
  6  7  8  9 10 11 12 12
$keyVarF
# Coupling of process variance parameters for log(F)-process (Fishing mortality
# normally applies to the first (fishing) fleet; therefore only first row is used)
  0  1  2  2  3  3  3  3
 -1 -1 -1 -1 -1 -1 -1 -1
 -1 -1 -1 -1 -1 -1 -1 -1
$keyVarLogN
# Coupling of the recruitment and survival process variance parameters for the
# log(N)-process at the different ages. It is advisable to have at least the first age
# class (recruitment) separate, because recruitment is a different process than
# survival.
 0 1 1 1 1 1 1 1
$keyVarObs
# Coupling of the variance parameters for the observations.
```

```
# First row refers to the coupling of the variance parameters for the catch data
# Second and further rows refers to coupling of the variance parameters for the indices
  0  1  1  1  1  1  2  2
  3  4  4  4  4  5  5 -1
  6  7  7  7  7  7  7  7
```

$obsCorStruct

```
# Covariance structure for each fleet ("ID" independent, "AR" AR(1), or "US" for unstructured). | Possible values are: "ID" "AR" "US"
 "ID" "AR" "ID"
```

$keyCorObs

```
# Coupling of correlation parameters can only be specified if the AR(1) structure is chosen above.
# NA's indicate where correlation parameters can be specified (-1 where they cannot).
#1-2 2-3 3-4 4-5 5-6 6-7 7-8
 NA  NA  NA  NA  NA  NA  NA
  0   0   0   0   0   0   0
 NA  NA  NA  NA  NA  NA  NA
```

$stockRecruitmentModelCode

```
# Stock recruitment code (0 for plain random walk).
 0
```

$fbarRange

```
# lowest and higest age included in Fbar
 2 6
```

$fixVarToWeight

```
# If weight attribute is supplied for observations this option sets the treatment (0 relative weight, 1 fix variance to weight). Can be specified fleetwise.
 0 0 0
```

$fracMixF

```
# The fraction of t(3) distribution used in logF increment distribution
 0
```

$fracMixN

```
# The fraction of t(3) distribution used in logN increment distribution (for each age group)
 0 0 0 0 0 0 0 0
```

$fracMixObs

```
# A vector with same length as number of fleets, where each element is the fraction of t(3) distribution used in the distribution of that fleet
 0 0 0
```

$stockWeightModel

```
# Integer code describing the treatment of stock weights in the model (0 use as known, 1 use as observations to inform stock weight process (GMRF with cohort and within year correlations)), 2 to add extra correlation to plusgroup
 1
```

$keyStockWeightMean

```
# Coupling of stock-weight process mean parameters (not used if stockWeightModel==0)
 0 1 2 3 4 5 6 7
```

$keyStockWeightObsVar

```
# Coupling of stock-weight observation variance parameters (not used if stockWeightModel==0)
 0 0 0 0 0 0 0 0
```

$catchWeightModel

```
# Integer code describing the treatment of catch weights in the model (0 use as known, 1 use as observations to inform catch weight process (GMRF with cohort and within year correlations)), 2 to add extra correlation to plusgroup
 1
```

$keyCatchWeightMean

```
# Coupling of catch-weight process mean parameters (not used if catchWeightModel==0)
  0 1 2 3 4 5 6 7
$keyCatchWeightObsVar
# Coupling of catch-weight observation variance parameters (not used if catchWeightModel==0)
  0 0 0 0 0 0 0 0
```

20.4 Assessment model results

The stock summary is given in Figure 20.9 and Table 20.12-14, while fishing mortality-at-age and abundance-at-age (stock numbers at age) estimated by the assessment model are presented in Tables 20.15 and 20.16, respectively.

20.4.1 Status of the stock

Fishing mortality in 2024 was 0.256 which is below F_{MSY} (0.386). From 2018-2021, Fbar (ages 2-6) was estimated slightly higher than F_{MSY}, but decreased to levels below F_{MSY} in 2022 and 2023 (Figure 20.10).

The SSB in 2024 was estimated at 4950 tonnes, which is 6.3% higher than the SSB in 2023 (Table 20.13). SSB has been below MSY $B_{trigger}$ (5046 tonnes) since 2022 but is predicted to increase to 5817 tonnes in 2025, above MSY $B_{trigger}$ in the short-term forecast (Figure 20.10, Table 20.13).

Recruitment in 2023 and 2024, has increased since the exceptionally low 2021 year, and was above the long-term mean.

20.4.2 Historic stock trends

SSB was at its highest in the early 1980s (possibly higher before that time for which no reliable data is available). From the 1980s up until the mid-1990s, SSB declined gradually while F increased (Figure 20.7). The lowest estimated value was estimated in 1995. SSB subsequently stayed around B_{lim} until 2007 when it rose to levels above MSY $B_{trigger}$. SSB dropped again to MSY $B_{trigger}$ in 2022 and has increased since. It is important to note that the SSB estimate before 1991 is based solely on landings data, as the BTS.COAST survey index only starts in 1991.

Recruitment has been variable over the time-series without a clear trend. The large recruitments in 2014 and 2015 have been well-above the long term mean and contributed to the increase in SSB in 2017 and 2018 while the lowest estimated recruitment in the time series was measured in 1992 at 3 605 740. During the years of missing landings-at-age data (1990s), recruitment values are estimated to have been at a lower than average level. The years 2021 was marked with low recruitment (below the long-term geometric mean of the time series), before showing an increase in 2022-2024.

Mean F peaked in 1990 at 0.68, but then declined to 0.39 in 1996, before rapidly increasing again to 0.66 in 2003. After 2003, there is a steep decline in F to 0.32 in 2007. Between 2008 and 2015, F fluctuated around 0.3. In the years following, F increased to 0.52 in 2021 but is estimated to be 0.26 in 2024. These trends correspond well with the trends in fishing effort of the beam trawl fleet.

20.4.3 Retrospective assessments

The results of the 5-year retrospective analysis are plotted in Figures 20.11. The retrospective plots for SSB, F and recruitment do not exhibit a strong negative or positive pattern. However, the retrospective pattern on F shows a tendency of overestimating F. Nevertheless, the Mohn's rho's associated with this retrospective analysis are within acceptable limits: -2% on SSB, -11% on F, and 25% on recruitment.

20.5 Model diagnostics

Model diagnostics are provided in Tables 20.17-24 and Figures 20.12-18.

The stability and estimability of a stock assessment model depends on the degree of collinearity between the parameters. When parameters are co-linear or correlated, the model can be sensitive to minor changes. A parameter correlation plot helps to identify the correlation between parameters. The correlation coefficient (varying between -1 and 1) is shown as a colour intensity as a function of the corresponding parameters. Ideally, the correlation between the parameters (except for a parameter with itself) should be 0, indicating the parameters are independent of each other. The parameter correlation plot for turbot shows some positive correlation between the catchability parameters (F_{par}), but no strong correlation between the other parameters (Figure 20.12).

To see how the SAM model has converged on the observation variances, the estimated observation variance (CV) of each data source in the assessment is plotted against the coefficient of variance of the estimate (Figure 20.13). Ideally all parameters should have a low CV. For turbot, the observation variance of the landing-at-age from age 2 onward is lowest, followed by BSAS. As such, the model assumes most information is available in these parameters giving them more weight in the assessment (Figure 20.14).

Survey catchabilities for all surveys with more than 1 age group are shown in Figure 20.15.

The estimated selectivity-at-age from 1981 to 2024 is shown in Figure 20.16. The selectivity at-age shows little trends in the past decade.

Residual plots of both observation and process errors show no substantial patterns (Figures 20.17 and 20.18).

The leave-one-out analysis currently cannot be run properly within both the stockassessment and FLSAM R packages due to the limited timeframe that the BSAS index has within the model. However, the leave-one out analysis can be partially run only showing the effects of removing catches or the BTS+COAST index (Figure 20.19). We see that the inclusion of the BTS+COAST index causes an increase in SSB and corresponding decrease in estimated fishing pressure.

20.6 Reference Points

Reference points were estimated during the benchmark. However, during WGNSSK 2025, changes in historical InterCatch data (2021-2023) and inconsistencies in the InterCatch age allocation matrix for 2023 led to a revision of reference points (Annex 8 Working Document 6). The reference point calculation used the same assumptions as during the benchmark.
The simulations were executed with the entire time-series of SR-pairs (1981–2024) and were run with 2000 iterations and applying a mixture of two SR-models, namely Segmented Regression and Ricker (sampling from 2000 fits) (Figure 20.20). Productivity and stock-recruit pairs over time are shown in Figures 20.21 and 20.22.

The table below shows the estimated reference points using the final WGNSSK 2025 assessment. Further details on the updated reference points can be found in the working document, ICES (2025).

Reference point	Estimate	Technical Basis
1. MSY $B_{trigger}$	5046	5th percentile of the SSB at MSY; in tonnes.
2. B_{pa}	5046	$B_{lim} \times \exp(1.645 \times \sigma)$, $\sigma = 0.20$; in tonnes.
3. B_{lim}	3631	Lowest SSB level that in the past has resulted in good recruitment; in tonnes.
4. F_{pa}=$F_{p0.05}$	0.386	The F that provides a 95% probability for SSB to be above B_{lim} ($F_{P.05}$ with advice rule).
5. F_{MSY}	0.386	Stochastic simulation (EqSim) based on the recruitment period 1981–2024. Capped at F_{pa}.
6. $F_{MSY\ lower}$	0.270	Consistent with ranges resulting in no more than 5% reduction in long-term yield compared with MSY.
7. $F_{MSY\ upper}$	0.386	Consistent with ranges resulting in no more than 5% reduction in long-term yield compared with MSY.

20.7 Short-term-forecast

The deterministic short-term forecast was implemented in FLR (Kell *et al.*, 2007) using the routines available in the Flasher package (https://flr-project.org/FLasher). Terminal year estimates from the SAM assessment were used as starting conditions. Since recruitment fluctuates over the time series (Figures 20.10 and 20.21), it was decided to set recruitment to follow a geometric mean for the entire time-series, including the latest estimate as the CV fell within acceptable limits.

Since stock and catch weight-at-age are modelled, we assume in the forecast that weights represent the average of the final three years of the time series. Maturity-at-age and time of spawning are fixed over time, and these values are used in the forecast. Selectivity-at-age is with minimal trends in recent years, but has changed in the past decade. Hence, a 5-year average was used for future years in the simulations.

The TAC has not been exhausted since 2016. In 2024, 87% of the single-species TAC was used. Using a TAC constraint assumption was, therefore, deemed inappropriate. Hence, the assumption for the intermediate year F used a status-quo F (F_{sq}) instead, scaled to the average $F_{ages\ 2\text{-}4}$ in 2024.

Assumptions made for the interim year and in the forecast. All weights are in tonnes, recruitment in thousands:

Variable	Value	Notes
$F_{ages\ 2–6}$ (2025)	0.26	F_{sq} = Average exploitation pattern (2020–2024), scaled to $F_{ages\ 2–6}$ 2024
SSB (2026)	6831	Short-term forecast (STF)
R_{age1} (2025, 2026)	8529	Geometric mean (GM, 1981–2024); thousands
Projected landings (2025)	2021	STF

Variable	Value	Notes
Catch (2025)	2148	(Projected landings)/(1–average discard rate); average discard rate by weight 2022–2024 = 5.9%.

The options table summarizes the outcomes of the short-term forecast. The numbers presented are the rounded values; actual calculations are performed with the exact numbers.

Basis	Total catch * (2026)	Projected landings (2026)	Projected discards ** (2026)	$F_{projected\ landings}$ (ages 2-6) (2026)	SSB (2027)	% SSB change ^	% advice change ^^
MSY approach: F_{MSY}	3292	3098	195	0.39	6426	-5.9	36
F_{MSY} upper	3292	3098	195	0.39	6426	-5.9	36
F_{MSY} lower	2465	2319	146	0.28	7189	5.2	1.61
F = 0	0	0	0	0	9521	39	-100
F_{pa} ($F_{p.05}$ with AR)	3292	3098	195	0.386	6426	-5.9	36
$F=F_{2025}$	2313	2176	137	0.26	7331	7.3	-4.7
SSB (2026) = B_{lim}	6464	6082	382	0.96	3631	-47	166
SSB (2026) = B_{pa}	4823	4538	285	0.63	5046	-26	99
SSB (2026) = MSY $B_{trigger}$	4823	4538	285	0.63	5046	-26	99
Rollover advice	2426	2283	143	0.27	7225	5.8	0
Multi-options table							
F = 0	0	0	0	0	9521	39	-100
F= 0.05	498	468	29	0.05	9044	32	-79
F = 0.10	972	914	57	0.10	8593	26	-60
F = 0.15	1424	1340	84	0.15	8164	19.5	-41
F = 0.20	1855	1746	110	0.20	7758	13.6	-24
F = 0.25	2266	2132	134	0.25	7374	8	-6.6
F = 0.30	2659	2502	157	0.30	7009	2.6	9.6
F = 0.35	3033	2854	179	0.35	6663	-2.4	25
F = 0.40	3391	3190	201	0.40	6335	-7.2	40
F = 0.45	3732	3511	221	0.45	6025	-11.8	54
F = 0.50	4058	3818	240	0.50	5730	-16.1	67

*** (projected landings) / (1 – average discard rate); average discard rate 2022–2024 = 5.9%**

**** Including BMS landings, assuming recent discard rate (5.9%).**

^ SSB 2027 relative to SSB 2026.

^^ Advice value for 2026 relative to advice value for 2025 (2426 t).

20.8 Management considerations

There are a number of EC regulations that affect the flatfish fisheries in the North Sea, e.g. as a basis for setting the TAC, limiting effort, and minimum mesh size. Since 2019, turbot falls under the landing obligation. The joint recommendation suggests a survivability exemption until 2028 for turbot caught by TBB gears with a cod end more than 80 mm in ICES Subarea 4 (Commission Delegated Regulation (EU) 2023/2459).

20.8.1 Effort regulations

The overall fleet capacity and deployed effort of the North Sea beam trawl fleet has been substantially reduced since 1995, due to a number of reasons, including the effort limitations for the recovery of the cod stock. In 2008, 25 vessels were decommissioned from the Dutch beam trawl fleet. In 2022, an increase in fuel prices resulted in reduced fishing effort, leading to a larger number of vessels remaining in the harbour. Additionally, the Dutch beam trawl fleet experienced substantial vessel decommissioning during the same year and fleet capacity has been greatly reduced since 2023.

20.8.2 Technical measures

Turbot is mainly taken by beam trawlers in a mixed fishery directed at sole and plaice in the southern and central part of the North Sea. Technical measures (EC Council Regulation 1543/2000) applicable to the mixed flatfish fishery affect turbot catches. The minimum mesh size of 80 mm in the beam trawl fishery selects sole at the minimum landing size (24 cm); however, this mesh size is likely to catch immature turbot (age 1 and 2 fish). Mesh enlargement would reduce the catch of smaller turbot, while at the same time potentially increasing the yield per recruit, but would also result in loss of marketable sole catches.

A closed area has been in operation since 1989 (the plaice box), and since 1995 this area has been closed in all quarters. The closed area applies to vessels using towed gears, but vessels smaller than 300 HP are exempted from the regulation. An additional technical measure concerning the fishing gear is the restriction of the aggregated beam length of beam trawlers to no more than 24 m. In the 12 nautical mile zone and in the plaice box, the maximum aggregated beam-length is 9 m.

20.8.3 Changes in TAC regulations

In 2023, the TAC for turbot and brill in the North Sea was combined as in previous years. In 2024, a regulatory amendment was introduced resulting in separate TAC allocations for turbot and brill to countries adjacent to the North Sea, effectively abolishing the combined TAC (EU–UK Joint Committee, 2023). This change marks and improvement in the management for turbot and brill in the North Sea as the previous combined TAC was deemed ineffective by ICES at reducing F due to species-specific changes in stock dynamics.

20.9 Industry Survey turbot and brill

The industry survey for turbot and brill (BSAS), set up by Dutch producer organisation VisNed and Wageningen Marine Research in 2018 to address limited catchability of (older) turbot in scientific beam trawl surveys, was incorporated into the assessment in the 2025 benchmark (ICES, 2025). Funding for the survey has not been secured post-2025. WGNSSK stresses the importance of the continuation of the industry survey given its inclusion in the assessment model. Termination of the survey would hamper ICES' ability to provide advice for this stock.

20.10 ACOM requests

In accordance to additional information requested by ACOM, Figures 20.23 -20.29 and Table 20.25 were included to add supplementary information on age distributions selectivity and how the current assessment compares with the one from the previous year.

The SSB-at-age plot suggests that the plus group is not as big as in previous years and that the overall increase seems to be driven by younger aged fish (Figure 20.23). Weights-at-age and selectivity do not show any obvious trends over the past 20 years (Figure 20.24). Figures 20.25-20.29 and Table 20.25 show differences between the current assessment and the previous years, however, as the stock went through benchmark earlier in the year which is why certain changes may appear drastic. More realistic assumptions in the benchmark model increased the overall natural mortality and caused an increase in recruitment throughout the time series (Figure 20.25). This shifted the age distribution in the assessment towards younger individuals (Figures 20.26-20.29). Furthermore, the SSB proportion within the plus group is now thought to have been overestimated in the previous model mostly due to the prior inclusion of an LPUE index which was removed at the benchmark.

20.11 Corrections to 2025 benchmark report

Upon review from the Benchmark Oversite Group (BOG), there were several minor issues in the North Sea turbot section of the 2025 Benchmark Workshop on Selected North Sea and Celtic Sea Stocks (WKBNSCS; ICES 2025a) detailed below:

In Table 2.7 of the benchmark report describing the different model runs trialled during the benchmark, the Gaussian Markov Random Field (GMRF) method for calculating weights-at-age was first used in model 2c. This method was selected by the group as the appropriate technique used for the assessment model. However, due to coding issues and lack of expert availability, models 3a through 6b used a modified von Bertalanffy method for quantifying weights-at-age until we were able to fix the GMRF issue later on in the benchmark process. Subsequently, the GMRF method was used for weights at age for models 7a through 9 (final model).

Regarding the new natural mortality vector, model 8 in Table 2.7 of the benchmark report uses an initial age varying natural mortality following the Lorenzen (1996) method. Upon finding a computational error in the initial calculation of mortality-at-age, the vector was updated for the final accepted model (model 9 in Table 2.7 of the Benchmark report).

In Table 2.9 of the benchmark report, which shows the reference points for North Sea turbot from 2018-2024, the F_{pa} value given is the $F_{P0.5}$ value without the advice rule and this value should be 0.856 which is $F_{P0.5}$ with the advice rule. The 2025 WGNSSK meeting has adjusted the reference points which are now different than those calculated from the benchmark report. The updated reference points can be found in section 20.6 of the current report.

Finally, a few typos were also found within the benchmark report and the caption for the Table 2.4 regarding AIC scores for the BSAS survey should specify that the same procedure was carried out for the other survey indices finding a similar pattern of lower AIC scores with the full GAM model with time varying and time invariant spatial components.

20.12 Issues for future benchmarks

20.12.1 Data

Estimates of discards are available (e.g. Dutch discards are available for 1999-present); however, age-length information is very limited. Age-information is based on a few fish sampled in the discards of some of the Danish and Belgian fleets (at-sea sampling). As a result, estimates of discards are highly uncertain, and not included in the current assessment. Future sampling effort needs to ensure a proper sampling coverage over the main fleets and countries for both landings and discards. Sampling should include age information for discards from all countries.

Coverage of age samples in landings of important fleet segments appears to be decreasing. Availability of age samples should be assessed to ensure adequate information on the age distribution in landings is available.

20.12.2 Assessment

No issues in the assessment relevant to future benchmarks were identified.

20.12.3 Short term forecast

The forecast is performed using future landings. Catch advice is derived by dividing the estimated landings with one minus the average discard rate observed in the previous three years.

Currently, a deterministic forecast is carried out. In the future, a stochastic forecast should be implemented to account for uncertainty in forward projections. This may be done in a revision for WGNSSK 2026.

20.13 References

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20.14 Tables and figures

Table 20.1. Turbot in Subarea 4. ICES estimated landings (tonnes) SOP corrected and discards of turbot.

Year	Landings	Landing SOP	Discards
1981	4755	1	
1982	4453	1	
1983	4575	1	
1984	5297	1	
1985	6188	1	
1986	5263	1	
1987	4271	1	
1988	4041	1	
1989	4927	1	
1990	5750	1	
1991	6340	-0.007	
1992	5933	-0.007	
1993	5546	-0.008	
1994	5244	-0.008	
1995	4671	-0.009	
1996	3644	-0.011	

Year	Landings	Landing SOP	Discards
1997	3382	-0.012	
1998	3086	1	
1999	3187	-0.012	
2000	4025	-0.009	
2001	4100	-0.009	
2002	3749	-0.010	
2003	3374	1	
2004	3317	1	
2005	3195	1	
2006	2976	1	
2007	3509	1	
2008	3005	1	
2009	3089	1	
2010	2692	1	
2011	2771	1	
2012	2914	1	
2013	2982	1	97
2014	2834	1	158
2015	2922	1	112
2016	3493	1	666
2017	3441	1	496
2018	3140	1	486
2019	3046	1	230
2020	3104	1	199
2021	2659	1	141
2022	1691	1	49
2023	1662	1	195
2024	1750	1	81

Table 20.2. Turbot in Subarea 4. Official estimated landings (tonnes) for each country participating in the fishery.

Year	Netherlands	UK	Denmark	Belgium	France	Germany	Norway	Other	Total
1975	3349	503	387	158	21	169	0	1	4588
1976	3253	631	588	146	38	156	0	2	4814
1977	2973	683	474	145	37	172	0	0	4484
1978	3196	752	693	170	50	173	0	< 1	5034
1979	3999	838	1164	187	22	151	0	3	6364
1980	3241	559	1360	162	17	146	0	0	5485
1981	3073	404	1044	142	6	86	0	0	4755
1982	3029	335	880	153	14	42	0	< 1	4453
1983	3163	277	893	174	24	44	0	< 1	4575
1984	3800*	282	886	242	40	46	0	1	5297
1985	4600*	312	983	222	37	34	0	0	6188
1986	3810*	287	997	133	5	31	0	0	5263
1987	2760*	345	988	130	21	27	0	0	4271
1988	2660	328	858	129	24	41	0	1	4041
1989	3666	333	637	176	30	85	0	0	4927
1990	3732	437	1046	292	52	184	0	7	5750
1991	3780	688	1233	350	64	186	30	9	6340
1992	3495	902	907	317	81	163	65	3	5933
1993	2939	1013	817	355	123	252	47	0	5546
1994	2724	882	862	330	141	263	42	0	5244
1995	2476	703	761	315	108	275	33	0	4671
1996	1776	687	618	210	160	157	36	0	3644
1997	1854	619	479	169	1	215	45	0	3382
1998	1695	582	392	198	22	164	33	< 1	3086
1999	1808	488	411	224	0	224	32	0	3187
2000	2280	549	469	302	21	349	55	0	4025
2001	2226	642	506	333	17	297	79	0	4100

Year	Netherlands	UK	Denmark	Belgium	France	Germany	Norway	Other	Total
2002	1898	551	677	243	15	280	85	0	3749
2003	1893	431	486	192	18	289	65	0	3374
2004	1762	463	518	207	15	278	74	0	3317
2005	1903	347	429	159	18	274	65	0	3195
2006	1828	381	338	147	22	221	40	< 1	2976
2007	2263	485	310	173	32	203	43	0	3508
2008	1744	370	457	182	21	199	32	< 1	3005
2009	1698	421	548	172	24	198	29	< 1	3090
2010	1469	386	467	118	37	191	26	< 1	2695
2011	1540	396	548	122	33	144	28	< 1	2811
2012	1740	366	481	150	30	187	37	< 1	2991
2013	1763	374	498	161	40	219	29	< 1	3085
2014	1594	391	433	176	42	197	38	< 1	2872
2015	1739	336	392	215	46	236	10	4	2978
2016	1854	404	505	339	38	273	8	1	3422
2017	2118	397	486	336	40	252	13	1	3641
2018	1914	368	331	267	27	306	15	1	3231
2019	1901	362	273	228	14	326	13	1	3121
2020	2099	354	257	161	5	297	17	1	3192
2021	1683	325	326	134	3	222	13	1	2705
2022	926	210	333	90	1	125	16	1	1702
2023*	828	238	255	125	< 1	172	28	1	1647
2024*	835	281	230	158	< 1	207	41	1	1754

* Preliminary

Table 20.3. Turbot in Subarea 4. Observed landings in numbers (units: thousands) SOP corrected.

Year	Age							
	1	2	3	4	5	6	7	8+
1981	0.000	297.536	751.303	529.395	455.757	174.143	66.672	106.476

Year	Age							
	1	2	3	4	5	6	7	8+
1982	0.000	168.924	1045.530	266.880	166.925	291.869	97.956	154.930
1983	0.000	392.746	657.507	467.973	107.468	110.399	175.856	198.327
1984	0.000	1244.974	1174.848	298.755	150.819	57.638	54.756	187.323
1985	0.000	732.041	2223.746	602.207	164.824	100.368	23.941	147.329
1986	0.000	353.580	1400.976	664.272	174.407	63.854	27.638	118.178
1987	12.262	611.207	514.999	637.618	149.029	49.048	17.921	66.025
1988	31.063	935.353	773.996	153.591	151.865	77.658	24.160	66.441
1989	0.000	641.708	1120.288	340.300	150.164	78.863	30.249	65.899
1990	42.868	954.063	1028.833	304.064	159.509	72.776	97.699	109.662
1991–1997	NO DATA							
1998	0.000	459.781	985.976	405.289	82.590	33.462	9.621	16.186
1999–2002	NO DATA							
2003	259.774	2363.264	570.141	367.770	87.564	40.766	25.588	25.394
2004	453.290	2063.262	825.674	144.077	85.892	10.067	7.850	6.326
2005	380.154	2191.337	797.918	254.655	27.424	24.160	2.873	21.221
2006	884.902	1645.164	807.535	119.124	35.110	7.900	16.176	18.132
2007	76.706	2715.900	601.933	278.406	39.361	28.417	8.064	15.543
2008	176.429	1342.581	816.641	218.982	194.120	46.856	12.814	10.165
2009	110.457	1016.694	949.607	410.079	86.929	24.472	10.772	18.104
2010	281.699	1419.201	390.191	312.867	173.683	89.102	30.930	19.771
2011	210.757	1940.194	602.162	110.621	137.555	76.954	32.224	23.577
2012	0.000	1911.605	777.988	267.077	42.511	63.987	73.107	24.751
2013	162.129	1484.665	1015.946	305.667	85.457	24.408	39.459	24.317
2014	68.896	391.794	650.547	683.844	137.556	121.935	38.029	105.115
2015	40.938	1265.013	483.799	339.712	329.270	114.230	44.945	82.995
2016	0.000	1125.497	1075.877	360.984	387.787	202.800	48.855	76.423
2017	5.507	263.098	1324.693	478.283	110.665	49.955	78.228	48.401
2018	187.291	733.128	494.694	948.980	263.545	71.155	47.308	74.676

Year	Age							
	1	2	3	4	5	6	7	8+
2019	172.159	1061.728	881.441	262.642	358.720	122.170	22.880	63.883
2020	209.328	1549.632	822.228	385.818	141.071	142.926	41.532	40.699
2021	0.000	652.895	929.355	483.655	182.535	62.664	66.885	59.057
2022	181.998	394.750	438.382	222.792	124.784	30.715	11.201	36.945
2023	99.565	553.734	398.675	228.952	140.761	73.281	25.148	27.836
2024	323.257	509.610	572.631	206.068	96.528	59.620	22.190	37.184

Table 20.4. Turbot in Subarea 4. Raw weights-at-age in the landings (units: kg).

Year	Age							
	1	2	3	4	5	6	7	8+
1981	NA	0.90	1.00	1.70	2.60	3.60	4.40	6.86
1982	NA	0.90	1.10	1.80	2.60	3.20	4.50	5.56
1983	NA	0.90	1.20	2.00	2.80	3.60	4.00	5.52
1984	NA	0.80	1.30	2.20	3.20	3.80	4.50	6.24
1985	NA	0.70	1.10	2.00	3.20	4.20	5.00	6.49
1986	NA	1.00	1.30	2.10	3.00	3.70	6.30	6.43
1987	0.70	1.10	1.60	2.10	3.80	4.60	6.10	7.96
1988	0.70	1.00	1.60	2.80	3.10	4.60	6.00	6.66
1989	NA	1.00	1.50	2.70	3.90	4.70	6.90	7.91
1990	0.90	1.00	1.60	2.70	3.40	5.40	5.60	7.37
1991–1997	NO DATA							
1998	NA	0.83	1.26	2.12	3.34	4.92	5.38	6.83
1999–2001	NO DATA							
2002	0.41	0.65	1.29	1.86	2.71	3.61	3.61	4.01
2003	0.5	0.62	1.15	1.78	2.24	2.74	2.59	4.98
2004	0.43	0.69	1.2	2.12	3.17	3.76	5.15	8.61
2005	0.44	0.62	1.13	1.89	2.89	3.47	4.6	4.96
2006	0.41	0.66	1.31	1.92	3.37	5.09	2.7	2.29

Year	Age							
	1	2	3	4	5	6	7	8+
2007	0.34	0.7	1.25	1.75	3.27	3.72	4.17	4.96
2008	0.37	0.68	1.27	1.78	1.79	2.76	4.91	4.21
2009	0.41	0.62	1.25	1.76	2.95	4.83	5.47	4.01
2010	0.35	0.61	1.07	1.62	2.19	2.67	2.65	4.51
2011	0.48	0.55	1.06	1.79	1.97	3.25	4.48	3.9
2012	NA	0.6	0.91	1.46	2.58	3.01	3.47	3.23
2013	0.61	0.61	1.00	1.64	2.23	3.41	2.27	2.91
2014	0.41	0.59	1.07	1.42	1.67	1.85	3.034	3.09
2015	0.41	0.59	1.1	1.3	1.67	2.12	2.78	3.16
2016	0.00	0.66	0.93	1.33	1.22	1.94	2.93	2.39
2017	0.54	0.98	1.18	1.74	2.15	2.37	3.07	3.51
2018	0.34	0.59	0.98	1.36	1.41	1.9	2.86	2.71
2019	0.44	0.58	0.94	1.5	1.77	2.11	3.63	2.24
2020	0.44	0.63	0.96	1.29	1.48	2.01	2.87	3.13
2021	0	0.57	0.97	1.39	1.57	1.66	2.69	2.4
2022	0.48	0.79	1.13	1.59	1.75	2.46	3.09	3.25
2023	0.37	0.62	1.02	1.5	1.78	1.78	2.8	3.6
2024	0.45	0.7	0.98	1.32	1.92	1.66	2.18	2.68

Table 20.5. Turbot in Subarea 4. Raw weights-at-age in the stock estimated as the catch weights in Q2 (units: kg).

Year	Age							
	1	2	3	4	5	6	7	8+
1981	NA	0.90	0.80	1.48	2.59	3.23	5.66	6.65
1982	NA	0.59	1.01	1.80	2.53	3.33	4.88	6.19
1983	NA	0.61	1.13	1.99	2.77	3.38	3.97	5.02
1984	NA	0.66	1.04	2.07	2.87	4.25	4.93	6.42
1985	NA	0.59	1.02	1.83	2.95	4.46	5.99	6.13
1986	NA	0.91	1.12	1.98	3.08	3.48	7.02	6.32

Year	Age							
	1	2	3	4	5	6	7	8+
1987	0.7	0.72	1.25	1.87	3.60	3.24	5.36	8.30
1988	0.7	1.16	1.65	2.65	3.31	5.78	7.24	8.05
1989	NA	0.81	1.48	2.96	5.30	5.77	8.26	8.31
1990	0.9	0.84	1.79	3.09	3.02	5.34	3.47	8.78
1991–1997	NO DATA							
1998	NA	0.8	1.03	1.67	3.08	5.06	2.57	7.49
1999–2001	NO DATA							
2002	0.44	0.60	1.28	1.94	2.64	3.68	2.95	4.05
2003	NA	0.50	1.14	1.99	2.45	2.82	4.14	3.43
2004	NA	0.52	1.10	1.90	2.47	2.91	5.35	6.44
2005	NA							
2006	NA							
2007	NA	0.59	1.10	1.57	2.58	2.71	1.72	4.88
2008	NA	0.65	1.14	1.44	2.10	5.16	6.01	7.13
2009	NA	0.44	0.80	1.51	1.65	3.55	4.70	4.97
2010	NA	0.45	1.04	1.62	2.30	2.38	2.71	5.42
2011	NA	0.39	0.95	1.88	2.01	4.00	4.42	5.27
2012	NA	0.51	0.85	1.42	2.20	2.67	2.58	4.18
2013	NA	0.59	0.95	1.60	2.18	3.30	2.51	4.25
2014	0.38	0.57	0.95	1.24	1.50	1.72	1.84	2.97
2015	0.41	0.49	0.89	0.93	1.46	1.40	1.37	4.46
2016	0.41	0.58	0.78	1.30	0.80	1.49	4.78	3.34
2017	0.39	0.38	0.92	1.60	2.04	2.31	2.87	3.24
2018	0.27	0.45	1.03	1.46	1.64	2.72	2.37	4.20
2019	0.44	0.39	0.86	1.37	2.04	2.25	4.25	3.20
2020	0.44	0.56	1.16	1.39	2.39	2.31	3.21	2.80
2021	NA	0.53	0.97	1.49	1.72	1.92	2.76	2.53
2022	0.48	0.74	1.06	1.59	2.01	2.45	2.91	2.61

Year	Age							
	1	2	3	4	5	6	7	8+
2023	0.3	0.63	1.09	1.44	1.76	1.83	2.90	3.42
2024	0.43	0.61	0.92	1.33	1.94	1.57	2.13	2.74

Table 20.6. Turbot in Subarea 4. Modelled weights-at-age in the catch (units: kg).

Year	Age							
	1	2	3	4	5	6	7	8+
1981	0.58	0.85	1.10	1.80	2.63	3.55	4.49	6.40
1982	0.58	0.86	1.19	1.86	2.65	3.42	4.48	5.72
1983	0.58	0.86	1.27	2.00	2.80	3.60	4.37	5.63
1984	0.59	0.82	1.30	2.11	3.02	3.82	4.64	5.86
1985	0.63	0.81	1.26	2.10	3.10	4.10	5.04	6.07
1986	0.67	0.95	1.36	2.16	3.14	4.09	5.68	6.22
1987	0.69	1.02	1.55	2.25	3.40	4.43	5.80	7.08
1988	0.70	1.01	1.60	2.56	3.33	4.59	5.94	7.09
1989	0.73	1.01	1.59	2.60	3.62	4.65	6.18	7.47
1990	0.78	1.02	1.61	2.60	3.51	4.88	5.82	7.31
1991	0.72	1.04	1.63	2.54	3.50	4.64	5.81	7.11
1992	0.68	1.02	1.63	2.52	3.45	4.53	5.70	6.98
1993	0.65	0.99	1.61	2.50	3.41	4.45	5.60	6.86
1994	0.62	0.95	1.57	2.46	3.38	4.39	5.51	6.75
1995	0.60	0.91	1.52	2.41	3.34	4.34	5.44	6.65
1996	0.58	0.88	1.46	2.35	3.29	4.30	5.38	6.57
1997	0.56	0.84	1.40	2.26	3.24	4.28	5.32	6.52
1998	0.54	0.82	1.32	2.15	3.15	4.34	5.23	6.50
1999	0.52	0.79	1.30	2.06	2.92	3.93	4.96	6.17
2000	0.49	0.76	1.27	2.00	2.77	3.63	4.59	5.79
2001	0.47	0.73	1.25	1.95	2.68	3.42	4.21	5.34
2002	0.44	0.69	1.24	1.90	2.63	3.36	3.82	4.83

Year	Age							
	1	2	3	4	5	6	7	8+
2003	0.47	0.67	1.19	1.91	2.54	3.19	3.60	4.53
2004	0.44	0.68	1.18	2.00	2.83	3.46	4.31	5.67
2005	0.44	0.66	1.17	1.94	2.90	3.52	4.13	5.29
2006	0.42	0.67	1.20	1.90	2.98	4.01	3.63	4.30
2007	0.39	0.68	1.19	1.80	2.81	3.70	4.24	4.73
2008	0.39	0.66	1.19	1.80	2.34	3.33	4.54	5.23
2009	0.40	0.62	1.15	1.75	2.52	3.56	4.49	5.17
2010	0.39	0.61	1.07	1.68	2.30	3.06	3.62	5.11
2011	0.44	0.58	1.02	1.68	2.17	3.02	3.92	4.76
2012	0.46	0.61	0.96	1.54	2.28	2.85	3.50	4.88
2013	0.50	0.62	0.98	1.51	2.06	2.82	2.91	4.56
2014	0.44	0.63	1.00	1.41	1.81	2.27	3.01	3.70
2015	0.44	0.62	1.02	1.35	1.70	2.20	2.89	3.51
2016	0.46	0.66	0.99	1.40	1.57	2.13	2.88	3.68
2017	0.47	0.74	1.05	1.49	1.81	2.20	2.86	3.53
2018	0.40	0.64	1.01	1.42	1.66	2.12	2.80	3.23
2019	0.43	0.60	0.96	1.42	1.71	2.12	2.96	2.91
2020	0.44	0.62	0.96	1.35	1.62	2.06	2.73	3.07
2021	0.45	0.62	0.98	1.39	1.66	1.96	2.66	2.82
2022	0.45	0.68	1.03	1.47	1.72	2.15	2.72	2.98
2023	0.41	0.64	1.01	1.47	1.77	1.96	2.61	3.00
2024	0.43	0.65	0.99	1.41	1.84	1.94	2.39	2.85

Table 20.7. Turbot in Subarea 4. Modelled weights-at-age in the stock (units: kg).

Year	Age							
	1	2	3	4	5	6	7	8+
1981	0.52	0.73	0.97	1.67	2.62	3.48	5.07	6.42
1982	0.52	0.65	1.09	1.78	2.60	3.51	4.64	6.18

Year	Age							
	1	2	3	4	5	6	7	8+
1983	0.54	0.64	1.14	1.91	2.75	3.65	4.43	5.76
1984	0.56	0.66	1.12	1.98	2.87	4.03	4.81	6.08
1985	0.61	0.68	1.14	1.96	2.94	4.20	5.37	6.28
1986	0.64	0.79	1.22	2.04	3.06	4.00	5.82	6.73
1987	0.69	0.82	1.36	2.13	3.34	4.12	5.63	7.57
1988	0.70	0.93	1.53	2.47	3.46	4.89	6.11	7.78
1989	0.72	0.88	1.57	2.65	3.90	4.97	6.27	8.01
1990	0.75	0.87	1.62	2.69	3.48	4.92	5.05	8.02
1991	0.69	0.87	1.55	2.55	3.48	4.62	5.31	7.26
1992	0.65	0.85	1.51	2.44	3.39	4.47	5.27	6.99
1993	0.62	0.81	1.46	2.36	3.28	4.33	5.16	6.81
1994	0.59	0.78	1.40	2.28	3.17	4.18	5.02	6.65
1995	0.58	0.75	1.35	2.20	3.07	4.03	4.86	6.48
1996	0.56	0.72	1.29	2.12	2.99	3.88	4.66	6.33
1997	0.55	0.71	1.23	2.02	2.95	3.77	4.34	6.22
1998	0.53	0.72	1.16	1.88	2.88	3.98	3.71	6.24
1999	0.52	0.69	1.20	1.89	2.70	3.64	3.98	5.46
2000	0.49	0.66	1.20	1.90	2.63	3.43	3.90	5.14
2001	0.47	0.64	1.19	1.90	2.61	3.35	3.72	4.83
2002	0.45	0.60	1.19	1.90	2.59	3.43	3.55	4.45
2003	0.44	0.56	1.12	1.90	2.54	3.26	4.02	4.35
2004	0.44	0.54	1.05	1.81	2.57	3.32	4.36	5.31
2005	0.44	0.54	1.00	1.63	2.54	3.72	4.21	5.33
2006	0.44	0.59	1.03	1.70	2.65	3.53	3.88	5.20
2007	0.43	0.58	1.04	1.61	2.51	3.34	3.30	5.16
2008	0.41	0.56	1.03	1.54	2.24	3.72	4.28	5.55
2009	0.39	0.50	0.94	1.56	2.03	3.24	4.16	5.27
2010	0.39	0.48	0.96	1.57	2.16	2.83	3.50	5.21

Year	Age							
	1	2	3	4	5	6	7	8+
2011	0.39	0.46	0.91	1.60	2.08	3.01	3.54	4.87
2012	0.40	0.50	0.87	1.45	2.06	2.67	3.00	4.37
2013	0.40	0.53	0.88	1.39	1.90	2.55	2.69	4.00
2014	0.39	0.53	0.88	1.25	1.63	2.05	2.37	3.49
2015	0.40	0.51	0.88	1.15	1.50	1.88	2.19	3.64
2016	0.39	0.51	0.86	1.29	1.35	1.89	2.90	3.37
2017	0.37	0.46	0.90	1.41	1.73	2.13	2.72	3.46
2018	0.35	0.47	0.92	1.43	1.78	2.40	2.65	3.63
2019	0.41	0.47	0.90	1.41	1.92	2.33	3.03	3.33
2020	0.43	0.53	0.97	1.41	1.98	2.32	2.85	3.15
2021	0.44	0.55	0.96	1.46	1.85	2.22	2.70	2.98
2022	0.44	0.61	0.99	1.48	1.86	2.27	2.69	2.99
2023	0.39	0.59	1.01	1.45	1.81	2.07	2.62	3.21
2024	0.41	0.56	0.96	1.43	1.87	2.00	2.40	3.07

Table 20.8. Turbot in Subarea 4. Natural mortality-at-age and maturity at age.

Age	1	2	3	4	5	6	7	8+
natural mortality	0.61	0.46	0.38	0.32	0.30	0.29	0.28	0.27
maturity	0	0.01	0.58	0.98	1	1	1	1

Table 20.9. Turbot in Subarea 4. Fraction of harvest before spawning and fraction of natural mortality before spawning.

Age	1	2	3	4	5	6	7	8+
Harvest	0	0	0	0	0	0	0	0
Natural mortality	0	0	0	0	0	0	0	0

Table 20.10. Turbot in Subarea 4. BTS+COAST survey index

	Age						
Year	1	2	3	4	5	6	7
1991	12.10	64.00	5.00	0.57	0.84	0.60	1.84

	Age						
Year	**1**	**2**	**3**	**4**	**5**	**6**	**7**
1992	16.20	38.00	11.90	0.17	0.71	0.55	0.73
1993	18.50	32.00	3.60	1.37	0.26	0.48	0.53
1994	26.40	46.00	2.40	0.30	0.85	0.25	0.32
1995	31.90	20.00	4.20	0.09	0.52	0.67	0.51
1996	7.00	44.00	2.00	0.78	0.54	0.36	0.97
1997	4.90	46.00	10.90	1.01	0.51	0.52	0.63
1998	41.90	29.00	8.40	5.37	0.72	NA	1.30
1999	35.60	40.00	7.00	3.22	0.29	0.13	0.18
2000	65.20	28.00	22.20	8.08	0.56	0.51	0.42
2001	20.60	56.00	4.10	4.60	0.17	0.11	3.22
2002	63.20	22.00	6.00	1.64	0.35	0.46	0.19
2003	64.40	44.00	2.80	2.21	0.81	0.76	0.76
2004	65.50	26.00	15.00	1.20	4.78	2.64	1.21
2005	36.70	65.00	26.20	5.29	1.98	NA	1.10
2006	80.60	42.00	13.40	3.83	1.80	3.38	0.28
2007	48.60	56.00	29.90	12.78	2.29	2.22	0.19
2008	38.60	59.00	27.10	6.75	3.70	0.72	4.10
2009	37.50	23.00	26.60	20.50	10.00	1.37	0.76
2010	39.10	30.00	7.00	1.74	2.76	0.99	0.30
2011	75.10	45.00	5.80	0.49	4.13	0.94	1.27
2012	37.50	57.00	26.90	12.92	2.16	3.20	6.29
2013	25.10	33.00	29.90	5.07	1.43	2.17	1.00
2014	71.50	17.00	19.70	14.32	1.85	1.34	0.32
2015	138.70	62.00	13.30	2.19	5.66	0.78	0.73
2016	34.30	96.00	32.90	0.32	1.16	1.29	1.52
2017	97.50	32.00	51.40	9.34	0.87	NA	1.58
2018	59.70	52.00	16.00	18.07	7.52	0.46	0.94
2019	126.00	50.00	27.70	1.35	5.44	0.20	1.02

Year	Age 1	2	3	4	5	6	7
2020	73.60	59.00	25.40	4.74	0.65	1.99	0.79
2021	38.30	32.00	20.90	9.61	1.88	NA	1.00
2022	54.50	22.00	19.70	4.83	1.36	0.56	0.39
2023	60.10	48.00	22.30	19.22	3.94	1.42	0.28
2024	89.20	35.00	41.70	4.64	2.99	2.69	0.64

Table 20.11. Turbot in Subarea 4. BSAS survey index

Year	Age 1	2	3	4	5	6	7	8
2019	24086	14658	4952	878	2439	644	258	190
2020	9655	12609	5408	1406	465	990	77	59
2021	7419	9053	4610	1975	690	103	303	118
2022	6682	5303	3619	1030	659	160	14	473
2023	13604	9488	2514	2005	661	335	63	31
2024	11661	9892	5343	1035	916	991	275	146

Table 20.12. Turbot in Subarea 4. Fbar (Ages 2–6) estimates from the assessment model.

Year	Low	Fbar	High
1981	0.203	0.261	0.336
1982	0.210	0.269	0.345
1983	0.243	0.311	0.398
1984	0.278	0.354	0.450
1985	0.328	0.419	0.535
1986	0.310	0.398	0.512
1987	0.274	0.348	0.443
1988	0.291	0.367	0.464
1989	0.368	0.467	0.593
1990	0.504	0.677	0.908
1991	0.457	0.638	0.889

Year	Low	Fbar	High
1992	0.403	0.574	0.819
1993	0.366	0.532	0.775
1994	0.328	0.474	0.686
1995	0.294	0.424	0.611
1996	0.273	0.390	0.557
1997	0.284	0.402	0.567
1998	0.400	0.522	0.682
1999	0.348	0.478	0.658
2000	0.336	0.474	0.669
2001	0.350	0.485	0.672
2002	0.399	0.551	0.759
2003	0.533	0.664	0.826
2004	0.392	0.494	0.621
2005	0.338	0.430	0.546
2006	0.259	0.330	0.421
2007	0.253	0.320	0.403
2008	0.281	0.355	0.448
2009	0.254	0.313	0.387
2010	0.295	0.362	0.443
2011	0.263	0.321	0.390
2012	0.238	0.288	0.350
2013	0.228	0.275	0.332
2014	0.271	0.326	0.393
2015	0.310	0.372	0.445
2016	0.375	0.456	0.554
2017	0.265	0.322	0.390
2018	0.357	0.431	0.519
2019	0.380	0.459	0.555
2020	0.411	0.503	0.617

Year	Low	Fbar	High
2021	0.411	0.524	0.669
2022	0.231	0.306	0.404
2023	0.217	0.293	0.396
2024	0.180	0.256	0.363
2025			

Table 20.13. Turbot in Subarea 4. Total and Spawning Stock Biomass estimates from the assessment model (tonnes).

Year	Low	TSB	High	Low	SSB	High
1981	16179	32810	42707	17299	24013	33333
1982	15092	30776	39793	15203	21078	29223
1983	15380	31806	40800	12915	17939	24919
1984	16454	31021	39156	11683	16214	22501
1985	16018	26745	33379	11669	15745	21246
1986	13779	24372	30433	10876	14635	19692
1987	12445	23482	29270	9393	12905	17729
1988	11819	23099	28679	9010	12326	16861
1989	12088	21006	26520	8925	12086	16367
1990	11567	18735	24895	7067	9788	13556
1991	10609	13188	19013	4058	6280	9720
1992	9919	10076	14786	3145	4838	7443
1993	9013	8772	12866	2490	3780	5740
1994	8089	8346	12202	2020	3015	4502
1995	7769	9787	14220	2014	2927	4256
1996	7539	10092	14053	2178	3066	4317
1997	7600	10124	13524	2881	3883	5234
1998	7677	11113	14957	3288	4290	5599
1999	7476	10489	14764	2642	3742	5300
2000	8087	11878	16802	2803	3962	5601
2001	7821	11016	14949	2820	3867	5302

Year	Low	TSB	High	Low	SSB	High
2002	7577	12213	15347	3285	4234	5459
2003	7394	12077	14918	3034	3776	4699
2004	7399	12656	15977	2697	3465	4451
2005	7464	12914	16224	2770	3653	4818
2006	25207	15540	19732	3289	4283	5576
2007	23802	15637	19258	3959	5110	6594
2008	24795	14114	17062	5081	6455	8199
2009	24576	13416	16328	5407	6860	8704
2010	21430	14413	17637	5210	6608	8380
2011	19519	16441	20437	4901	6220	7895
2012	18838	16195	19515	5062	6274	7777
2013	18605	14864	17542	5953	7223	8765
2014	16638	15516	18353	6081	7307	8779
2015	14099	16643	19929	5031	6033	7234
2016	9147	14529	17030	5163	6142	7307
2017	6866	14149	16639	6171	7381	8827
2018	5981	14433	16971	6212	7494	9042
2019	5708	15051	18062	5130	6140	7349
2020	6736	13816	16534	4709	5671	6829
2021	7248	11788	14365	4400	5424	6687
2022	7579	11308	14290	3686	4714	6028
2023	8256	11801	15384	3547	4658	6116
2024	7451	13623	19199	3601	4950	6804
2025		13784			5817	

Table 20.14. Turbot in Subarea 4. Recruitment (Age 1) estimates from the assessment model (units: thousands).

Year	Low	Value	High
1981	5129.52	7120.78	9885.06
1982	8054.86	10911.33	14780.79

Year	Low	Value	High
1983	12223.01	16663.74	22717.84
1984	9024.85	12324.58	16830.78
1985	4690.70	6392.44	8711.54
1986	5882.46	7933.55	10699.83
1987	6463.31	8752.74	11853.13
1988	4941.57	6876.47	9568.99
1989	3718.52	5689.47	8705.11
1990	3896.91	6754.68	11708.16
1991	2387.34	4082.06	6979.84
1992	2130.42	3605.74	6102.71
1993	2805.06	4617.84	7602.16
1994	3037.79	4833.99	7692.27
1995	4839.00	7447.40	11461.83
1996	4032.35	5952.94	8788.29
1997	3287.93	4923.62	7373.05
1998	4743.80	7435.80	11655.45
1999	3930.13	6258.49	9966.24
2000	6232.75	9602.19	14793.16
2001	4319.78	6433.19	9580.55
2002	8104.37	10739.96	14232.67
2003	7729.92	10228.74	13535.33
2004	9638.55	12599.51	16470.09
2005	8507.48	11142.91	14594.74
2006	11815.42	15536.17	20428.60
2007	8321.57	11023.22	14601.97
2008	5003.79	6816.66	9286.33
2009	7271.58	9543.06	12524.10
2010	9475.18	12370.68	16151.02
2011	11937.56	16085.16	21673.80

Year	Low	Value	High
2012	8426.21	11071.17	14546.37
2013	5300.62	7095.91	9499.26
2014	10186.95	13392.97	17608.00
2015	11443.82	15539.58	21101.22
2016	4618.70	6424.40	8936.03
2017	6854.27	9038.79	11919.54
2018	8615.79	11308.73	14843.37
2019	9386.64	12703.76	17193.10
2020	6172.43	8177.40	10833.63
2021	4780.03	6508.79	8862.77
2022	5959.58	8467.04	12029.49
2023	6376.50	9514.08	14195.51
2024	6845.25	11822.41	20418.46
2025*		8529	

Table 20.15. Turbot in Subarea 4. Fishing mortality-at-age as estimated by the assessment model.

Year	Age							
	1	2	3	4	5	6	7	8+
1981	0.001	0.069	0.427	0.384	0.221	0.204	0.155	0.155
1982	0.001	0.069	0.422	0.389	0.238	0.228	0.178	0.178
1983	0.001	0.092	0.468	0.431	0.290	0.272	0.209	0.209
1984	0.002	0.134	0.540	0.473	0.330	0.293	0.209	0.209
1985	0.002	0.160	0.609	0.544	0.428	0.354	0.229	0.229
1986	0.002	0.163	0.583	0.520	0.400	0.326	0.214	0.214
1987	0.003	0.175	0.557	0.461	0.307	0.242	0.171	0.171
1988	0.004	0.219	0.600	0.457	0.314	0.248	0.183	0.183
1989	0.006	0.281	0.731	0.557	0.448	0.318	0.233	0.233
1990	0.012	0.431	0.953	0.720	0.725	0.553	0.420	0.420
1991	0.011	0.400	0.927	0.701	0.659	0.501	0.390	0.390

Year	Age							
	1	2	3	4	5	6	7	8+
1992	0.010	0.362	0.858	0.656	0.566	0.430	0.345	0.345
1993	0.009	0.338	0.817	0.622	0.505	0.380	0.309	0.309
1994	0.007	0.291	0.747	0.575	0.433	0.325	0.269	0.269
1995	0.006	0.249	0.672	0.532	0.381	0.285	0.239	0.239
1996	0.005	0.213	0.613	0.501	0.354	0.267	0.226	0.226
1997	0.005	0.208	0.602	0.514	0.392	0.292	0.245	0.245
1998	0.008	0.269	0.703	0.625	0.590	0.425	0.342	0.342
1999	0.009	0.275	0.653	0.579	0.522	0.364	0.302	0.302
2000	0.012	0.323	0.658	0.565	0.494	0.330	0.280	0.280
2001	0.017	0.382	0.681	0.570	0.480	0.312	0.270	0.270
2002	0.028	0.508	0.747	0.618	0.544	0.336	0.296	0.296
2003	0.046	0.671	0.843	0.696	0.689	0.418	0.367	0.367
2004	0.044	0.613	0.711	0.547	0.377	0.220	0.182	0.182
2005	0.042	0.559	0.633	0.472	0.289	0.195	0.179	0.179
2006	0.030	0.439	0.491	0.361	0.198	0.160	0.173	0.173
2007	0.022	0.395	0.445	0.345	0.220	0.193	0.185	0.185
2008	0.023	0.388	0.445	0.370	0.313	0.257	0.205	0.205
2009	0.023	0.400	0.429	0.333	0.223	0.181	0.160	0.160
2010	0.026	0.426	0.456	0.369	0.289	0.269	0.233	0.233
2011	0.021	0.373	0.414	0.341	0.245	0.230	0.212	0.212
2012	0.016	0.305	0.380	0.326	0.216	0.214	0.208	0.208
2013	0.014	0.271	0.361	0.326	0.211	0.206	0.195	0.195
2014	0.008	0.200	0.355	0.373	0.331	0.372	0.360	0.360
2015	0.007	0.194	0.376	0.417	0.432	0.439	0.412	0.412
2016	0.007	0.188	0.413	0.500	0.636	0.543	0.493	0.493
2017	0.004	0.127	0.334	0.401	0.398	0.349	0.350	0.350
2018	0.011	0.206	0.428	0.494	0.545	0.481	0.448	0.448
2019	0.019	0.249	0.461	0.515	0.569	0.503	0.440	0.440

Year	Age							
	1	2	3	4	5	6	7	8+
2020	0.032	0.295	0.488	0.540	0.651	0.541	0.463	0.463
2021	0.032	0.264	0.470	0.528	0.704	0.656	0.554	0.554
2022	0.018	0.165	0.333	0.354	0.342	0.334	0.336	0.336
2023	0.018	0.157	0.320	0.338	0.320	0.332	0.351	0.351
2024	0.019	0.147	0.302	0.311	0.258	0.261	0.289	0.289

Table 20.16. Turbot in Subarea 4. Population abundance (stock numbers at age) as estimated by the assessment model (units: thousands).

Year	Age							
	1	2	3	4	5	6	7	8+
1981	7120.78	5648.69	2303.66	1884.51	2705.92	1095.74	556.22	915.72
1982	10911.33	3777.62	3331.35	1005.96	924.86	1612.69	673.86	971.42
1983	16663.74	5935.25	2254.46	1478.46	487.02	539.11	966.29	1051.61
1984	12324.58	9449.96	3454.55	1001.67	679.89	265.43	309.85	1224.39
1985	6392.44	6835.02	5104.65	1410.94	477.19	361.35	147.39	937.29
1986	7933.55	3298.93	3830.58	1818.49	584.70	236.16	187.91	658.40
1987	8752.74	4336.78	1722.69	1539.62	741.24	283.65	128.44	518.04
1988	6876.47	4875.94	2261.65	689.84	696.36	393.87	166.91	417.03
1989	5689.47	3685.58	2423.20	861.87	349.40	379.10	224.18	368.92
1990	6754.68	3013.71	1782.72	777.65	356.87	170.05	215.48	357.24
1991	4082.06	3802.11	1177.83	469.83	274.69	128.01	73.46	285.62
1992	3605.74	2191.35	1649.41	306.21	171.35	105.39	58.19	183.88
1993	4617.84	1910.71	944.30	487.96	114.37	72.56	51.43	129.95
1994	4833.99	2541.47	830.58	283.46	193.15	51.33	37.29	101.17
1995	7447.40	2547.92	1200.38	264.39	117.02	93.92	27.94	81.02
1996	5952.94	4200.15	1221.24	423.37	115.39	59.55	53.41	65.80
1997	4923.62	3382.11	2202.12	454.65	189.03	61.23	34.38	72.84
1998	7435.80	2723.41	1788.17	855.62	198.80	97.07	34.80	64.29
1999	6258.49	4132.38	1370.70	597.10	326.25	81.34	47.62	53.46

Year	Age							
	1	2	3	4	5	6	7	8+
2000	9602.19	3241.67	2030.98	513.93	234.89	145.05	42.65	57.15
2001	6433.19	5351.71	1422.38	734.33	215.11	104.19	79.39	57.95
2002	10739.96	3313.60	2361.71	483.43	300.31	103.48	57.25	79.56
2003	10228.74	5825.81	1208.68	801.18	183.56	131.93	58.06	78.33
2004	12599.51	5212.50	1908.78	351.77	298.64	64.70	66.04	70.89
2005	11142.91	6640.48	1736.91	649.45	142.30	151.16	36.99	92.34
2006	15536.17	5728.57	2440.40	588.65	280.95	76.64	92.60	83.54
2007	11023.22	8347.31	2295.74	1048.22	289.91	171.70	48.71	107.90
2008	6816.66	6025.79	3455.95	1018.39	532.57	175.93	103.79	95.19
2009	9543.06	3409.70	2732.30	1567.02	535.03	256.01	98.18	121.25
2010	12370.68	5166.88	1325.65	1242.01	809.81	328.91	156.74	135.25
2011	16085.16	6621.09	2184.66	537.81	653.80	449.37	184.70	168.58
2012	11071.17	8923.01	2807.76	1043.31	277.68	383.31	274.44	206.74
2013	7095.91	6017.01	4330.21	1271.15	554.02	168.58	234.77	281.03
2014	13392.97	3552.44	2795.15	2204.91	639.95	347.45	106.07	327.85
2015	15539.58	7409.76	1845.66	1324.06	1120.74	340.40	177.44	234.62
2016	6424.40	8833.12	3793.33	874.67	653.95	548.74	163.44	209.94
2017	9038.79	3223.58	4923.95	1683.89	387.75	245.86	245.21	174.94
2018	11308.73	4937.95	1726.44	2500.05	792.77	197.10	132.14	226.39
2019	12703.76	6024.66	2551.00	746.64	1099.90	331.32	92.86	175.74
2020	8177.40	6531.17	2925.04	1068.21	317.44	460.40	146.72	128.79
2021	6508.79	4154.98	2862.65	1247.77	432.24	119.76	205.16	134.80
2022	8467.04	3316.28	2013.61	1142.47	516.04	151.69	44.37	151.33
2023	9514.08	4648.50	1757.22	1006.47	584.29	270.69	83.45	104.01
2024	11822.41	4942.36	2563.39	875.35	515.34	318.86	143.91	105.95

Table 20.17. Turbot in Subarea 4. Predicted catch numbers-at-age (units: thousands).

Year	Age							
	1	2	3	4	5	6	7	8+
1981	2.99	304.02	675.02	518.89	465.29	176.60	70.03	115.48
1982	4.57	202.90	967.48	279.69	170.12	287.38	96.35	139.12
1983	11.07	417.26	711.95	447.29	106.71	112.57	160.09	174.51
1984	15.27	950.78	1221.01	326.77	166.21	59.05	51.29	203.01
1985	10.63	811.98	1975.37	513.47	144.88	94.42	26.54	169.08
1986	13.59	398.79	1435.38	638.80	168.16	57.53	31.86	111.82
1987	16.92	560.95	623.32	492.21	170.28	53.43	17.75	71.71
1988	20.49	772.51	865.27	218.72	163.32	75.74	24.56	61.46
1989	25.10	729.45	1070.71	319.10	110.17	90.53	40.97	67.52
1990	59.18	858.85	939.15	347.77	161.75	63.68	65.15	108.18
1991	33.29	1019.34	609.65	206.07	116.36	44.39	20.91	81.44
1992	26.14	539.82	812.60	128.12	64.83	32.37	14.94	47.29
1993	31.32	443.90	450.25	196.37	39.64	20.12	12.04	30.47
1994	26.95	519.49	372.33	107.61	59.21	12.48	7.73	21.01
1995	33.55	453.27	499.62	94.51	32.29	20.41	5.22	15.17
1996	21.81	650.12	475.02	144.49	29.99	12.21	9.48	11.70
1997	18.25	512.73	845.21	158.32	53.39	13.59	6.58	13.96
1998	44.01	518.63	768.39	345.35	77.57	29.55	8.88	16.43
1999	40.46	802.66	558.47	227.74	115.97	21.80	10.93	12.29
2000	85.10	725.48	832.44	192.53	80.01	35.77	9.15	12.28
2001	79.25	1381.21	597.59	276.77	71.60	24.48	16.55	12.10
2002	220.20	1077.15	1058.82	193.68	110.21	25.89	12.90	17.96
2003	347.58	2338.21	588.46	349.81	80.28	39.62	15.71	21.23
2004	410.23	1956.37	827.19	128.52	81.74	11.16	9.65	10.38
2005	342.32	2323.24	691.62	211.58	31.08	23.41	5.31	13.27
2006	343.39	1657.63	801.03	153.74	43.86	9.92	12.90	11.66
2007	181.33	2214.36	696.68	263.95	49.66	26.28	7.22	16.01
2008	115.10	1573.91	1048.77	271.50	124.59	34.92	16.90	15.53

Year	Age							
	1	2	3	4	5	6	7	8+
2009	162.48	914.24	804.29	382.95	92.73	36.97	12.77	15.80
2010	242.48	1459.22	409.81	330.49	176.92	67.84	28.61	24.73
2011	255.05	1672.55	624.82	134.07	123.48	80.80	30.98	28.32
2012	133.74	1900.33	748.16	250.26	46.84	64.49	45.21	34.11
2013	74.83	1156.43	1105.57	304.47	91.57	27.44	36.51	43.78
2014	76.43	519.09	704.02	591.90	157.17	94.64	28.26	87.50
2015	77.32	1053.78	487.61	390.08	343.31	106.31	52.87	70.02
2016	31.86	1221.97	1082.65	297.94	269.85	202.64	56.20	72.30
2017	24.31	308.39	1177.06	480.16	111.08	63.46	63.86	45.63
2018	95.96	742.02	506.93	843.37	291.49	66.25	42.13	72.29
2019	180.93	1072.65	795.68	260.26	418.10	115.29	29.17	55.29
2020	191.40	1350.00	955.43	386.63	133.28	169.45	48.06	42.25
2021	154.29	778.58	907.38	443.53	191.82	50.88	77.19	50.80
2022	112.77	407.22	480.00	293.77	130.05	37.75	11.15	38.10
2023	127.00	543.07	404.75	248.65	139.12	67.04	21.76	27.17
2024	163.77	545.25	561.23	201.38	101.80	64.18	31.79	23.44

Table 20.18. Turbot in Subarea 4. Catch-at-age residuals.

Year	Age							
	1	2	3	4	5	6	7	8+
1981	0.000	-0.055	0.458	1.440	2.035	0.357	2.862	0.000
1982	0.000	0.010	0.507	-1.262	-1.341	-2.094	-0.241	0.185
1983	0.000	1.447	-0.042	0.453	0.144	-0.151	0.085	0.025
1984	0.000	2.193	-0.198	0.249	-0.434	-0.276	-0.177	-0.914
1985	0.000	-0.517	-0.048	0.970	1.295	-0.106	-0.762	-0.830
1986	0.000	-1.316	-0.347	-0.808	0.062	0.369	-0.570	-0.068
1987	0.303	0.716	-1.845	0.781	-1.456	-0.939	0.144	-0.497
1988	0.783	0.782	-1.091	-1.338	0.268	0.028	0.215	0.049

Year	Age							
	1	2	3	4	5	6	7	8+
1989	0.000	-0.664	0.169	0.589	2.544	-0.553	-0.658	-0.306
1990	0.485	0.355	0.134	-1.029	0.425	1.603	2.068	-0.286
1991–1997	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1998	0.000	1.511	2.068	2.240	1.339	1.086	0.389	-0.285
1999–2002	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2003	1.478	0.989	-1.157	1.203	-0.135	0.953	2.056	0.390
2004	0.660	-0.456	-0.777	-0.019	-0.286	-1.682	-1.218	-1.706
2005	0.248	-0.283	-0.626	0.221	-1.063	0.442	-1.747	2.205
2006	1.047	-0.627	-0.716	-2.215	-1.017	-0.170	1.105	1.326
2007	-1.623	0.956	-0.809	1.048	0.169	1.297	-0.190	-0.715
2008	-0.023	-0.481	-0.793	0.183	2.806	1.000	-1.781	-1.660
2009	-0.388	-0.457	1.305	0.505	-0.433	-2.715	-0.268	0.570
2010	0.817	0.453	-1.215	0.340	0.505	2.277	0.118	-0.966
2011	-0.106	0.755	0.384	-1.214	0.478	-0.253	0.094	-1.015
2012	0.000	-0.141	0.343	1.331	-0.809	0.337	1.614	-1.736
2013	-0.148	-0.038	0.699	0.155	-0.209	-0.085	0.288	-2.115
2014	-0.691	-2.168	0.503	2.327	0.447	2.385	1.458	0.695
2015	-0.671	1.364	0.857	-0.105	0.657	-0.148	-0.710	0.714
2016	0.000	-0.402	0.718	1.455	1.922	-0.642	-0.497	0.332
2017	-2.115	-1.816	1.347	-0.433	-0.552	-1.376	1.096	0.279
2018	2.599	0.106	-0.553	0.756	-0.697	0.945	0.416	0.230
2019	0.953	-0.138	0.186	-0.632	-0.542	-0.130	-0.676	0.752
2020	0.493	-0.105	-0.908	-0.329	0.570	-0.560	-0.481	-0.037
2021	0.000	-1.214	-0.596	0.384	0.076	0.886	-0.171	0.914
2022	0.722	-1.434	-1.075	-2.525	-0.855	-1.601	-0.403	0.249
2023	0.059	0.131	-0.217	-0.222	-0.035	0.460	0.788	-0.325
2024	0.988	-1.130	0.157	-0.147	-0.847	-0.478	-0.989	1.722

Table 20.19. Turbot in Subarea 4. Predicted index-at-age BTS+COAST.

	Age						
Year	**1**	**2**	**3**	**4**	**5**	**6**	**7**
1991	19.582	31.513	5.642	1.405	0.907	0.451	1.381
1992	17.312	18.663	8.290	0.945	0.604	0.391	0.961
1993	22.182	16.552	4.886	1.543	0.421	0.279	0.738
1994	23.246	22.751	4.517	0.926	0.748	0.205	0.580
1995	35.850	23.501	6.879	0.891	0.470	0.385	0.466
1996	28.679	39.728	7.297	1.458	0.472	0.248	0.515
1997	23.719	32.100	13.259	1.551	0.754	0.250	0.457
1998	35.747	24.770	10.028	2.700	0.690	0.361	0.394
1999	30.071	37.424	7.963	1.946	1.187	0.316	0.413
2000	46.032	28.369	11.755	1.691	0.872	0.577	0.415
2001	30.739	44.920	8.100	2.409	0.806	0.420	0.574
2002	50.916	25.454	12.842	1.532	1.076	0.410	0.562
2003	47.861	39.892	6.140	2.404	0.593	0.493	0.533
2004	59.036	37.192	10.641	1.173	1.203	0.278	0.609
2005	52.305	49.229	10.235	2.282	0.610	0.661	0.577
2006	73.540	46.192	15.888	2.238	1.284	0.343	0.789
2007	52.463	69.433	15.439	4.028	1.305	0.752	0.696
2008	32.429	50.382	23.241	3.847	2.245	0.736	0.871
2009	45.393	28.260	18.589	6.073	2.403	1.131	0.992
2010	58.698	42.043	8.850	4.695	3.471	1.366	1.254
2011	76.598	55.955	15.019	2.073	2.891	1.917	1.539
2012	52.912	79.089	19.771	4.064	1.253	1.655	2.103
2013	33.963	54.618	30.900	4.953	2.509	0.732	2.275
2014	64.398	33.913	20.028	8.312	2.662	1.342	1.704
2015	74.771	71.030	13.032	4.837	4.343	1.254	1.560
2016	30.913	85.019	26.098	3.014	2.195	1.878	1.335
2017	43.586	32.405	35.813	6.223	1.539	0.965	1.660
2018	54.233	46.933	11.756	8.653	2.837	0.705	1.323

Year	Age 1	2	3	4	5	6	7
2019	60.589	55.560	16.970	2.546	3.870	1.166	0.997
2020	38.659	58.311	19.084	3.577	1.054	1.578	1.005
2021	30.761	37.921	18.917	4.216	1.383	0.379	1.164
2022	40.418	32.438	14.657	4.364	2.132	0.602	0.782
2023	45.415	45.747	12.909	3.889	2.451	1.075	0.741
2024	56.407	48.960	19.076	3.447	2.258	1.331	1.031

Table 20.20. Turbot in Subarea 4. Index-at-age residuals BTS+COAST

Year	Age						
	1	2	3	4	5	6	7
1991	-1.684	1.107	-1.327	-1.540	0.065	0.266	-0.051
1992	-0.271	0.995	0.445	-2.710	0.965	0.392	-0.617
1993	0.001	0.803	-0.569	0.260	-0.372	0.910	-0.550
1994	0.452	1.109	-1.111	-1.095	1.025	0.463	-0.624
1995	0.448	-0.186	-0.231	-2.449	1.484	1.027	0.178
1996	-1.705	1.393	-1.201	0.379	1.157	0.668	0.712
1997	-1.291	2.108	0.679	0.038	0.063	1.150	0.128
1998	1.425	0.554	0.121	1.245	0.029	0.000	1.459
1999	0.479	0.481	0.316	0.976	-1.894	-0.377	-0.591
2000	1.008	-0.463	1.366	2.427	-0.744	0.425	0.307
2001	-1.003	0.521	-1.368	1.214	-1.906	-0.533	2.558
2002	0.999	-0.761	-0.776	0.230	-0.955	1.177	-0.988
2003	0.201	-0.007	-0.945	0.557	0.628	0.859	0.773
2004	0.048	-0.703	0.902	0.324	2.539	2.290	0.633
2005	-0.799	0.553	1.166	1.034	1.430	0.000	0.365
2006	0.495	0.022	0.174	0.874	0.180	2.125	-2.125
2007	-0.168	-0.183	1.224	1.445	0.140	0.573	-2.181
2008	-0.155	0.624	0.279	0.659	-0.168	-0.551	1.336

Year	Age						
	1	2	3	4	5	6	7
2009	-0.243	-0.799	0.579	1.540	1.571	-0.698	-1.043
2010	-0.173	-0.243	-0.615	-1.493	-0.430	-0.719	-2.127
2011	0.734	-0.046	-0.923	-1.792	1.027	-0.929	-0.511
2012	-0.537	0.375	0.769	1.820	0.601	0.526	0.777
2013	-1.050	-0.271	0.528	-0.070	-0.774	1.267	-1.393
2014	1.592	-0.761	0.062	0.677	-0.995	-0.136	-1.961
2015	1.736	-0.183	0.109	-1.206	0.374	-0.667	-0.717
2016	-0.835	0.754	0.239	-3.347	-0.327	0.135	0.487
2017	1.766	-0.584	0.579	0.438	-0.684	0.000	0.211
2018	-0.334	-0.039	0.196	0.942	1.104	-0.709	-0.143
2019	0.573	-0.734	0.420	-1.242	0.506	-2.150	0.556
2020	-0.477	-0.639	0.219	-0.008	-1.170	0.164	-0.653
2021	-0.328	-0.561	-0.279	0.948	-0.196	0.000	-0.266
2022	0.400	-0.725	0.528	-0.019	-0.864	-0.206	-1.005
2023	0.834	0.204	0.902	2.364	0.437	0.194	-1.234

Table 20.21. Turbot in Subarea 4. Predicted index-at-age BSAS

	Age							
Year	1	2	3	4	5	6	7	8
2019	15111.096	11798.074	4301.780	921.493	1572.941	491.547	85.335	161.911
2020	9641.542	12382.168	4837.832	1294.723	428.528	665.107	132.610	116.701
2021	7671.957	8052.374	4795.519	1525.784	562.360	159.553	174.004	114.622
2022	10080.427	6888.061	3715.437	1579.326	866.481	253.568	43.865	149.999
2023	11326.650	9714.158	3272.446	1407.456	996.426	453.082	81.641	102.015
2024	14067.974	10396.577	4835.669	1247.602	917.934	560.972	147.075	108.551

Table 20.22. Turbot in Subarea 4. Index-at-age residuals BSAS

	Age							
Year	1	2	3	4	5	6	7	8
2019	0.016	-0.142	-0.109	-0.257	0.791	0.415	2.110	-0.162

	Age							
Year	1	2	3	4	5	6	7	8
2020	-1.777	-0.574	0.025	-0.194	-0.253	0.420	-1.512	-1.519
2021	-0.819	-0.126	-0.704	0.125	-0.220	-1.435	1.087	-0.072
2022	-1.168	-0.463	0.065	-0.842	-0.573	-0.993	-2.369	2.327
2023	0.451	0.097	-0.481	0.750	-0.731	-0.529	-0.335	-2.176
2024	-0.864	-0.086	0.259	-0.291	0.195	1.436	1.477	0.654

Table 20.23. Turbot in Subarea 4. Fit parameters.

Name	value	std.dev
logFpar	-4.903	0.105
logFpar	-4.184	0.129
logFpar	-4.422	0.131
logFpar	-5.09	0.137
logFpar	-5.036	0.147
logFpar	-5.093	0.18
logFpar	0.616	0.178
logFpar	1.174	0.226
logFpar	1.114	0.226
logFpar	0.801	0.23
logFpar	0.972	0.239
logFpar	0.95	0.259
logFpar	0.42	0.28
logSdLogFsta	-0.517	0.339
logSdLogFsta	-1.186	0.187
logSdLogFsta	-1.735	0.2
logSdLogFsta	-1.12	0.168
logSdLogN	-0.899	0.17
logSdLogN	-2.321	0.396
logSdLogObs	-0.354	0.24

Name	value	std.dev
logSdLogObs	-1.577	0.14
logSdLogObs	-1.159	0.145
logSdLogObs	-0.637	0.145
logSdLogObs	-0.368	0.069
logSdLogObs	-0.129	0.095
logSdLogObs	-1.151	0.367
logSdLogObs	-0.712	0.123
transfIRARdist	1.058	0.303

Table 20.24. Turbot in Subarea 4. Negative Log-Likelihood.

364.8018

Table 20.25. Comparison between the WGNSSK 2025 and 2024 estimates for recruitment, catch, fishing pressure (F) and targeted fishing pressure for total allowable catch (Target F for TAC). Note that values are not comparable due to the 2025 benchmark.

Parameter	Year*	Current Assessment (2025)	Previous assessment (2024)
Assumed recruitment	2024	11822 thousand	4370 thousand
	2025	8259 thousand	4372 thousand
Catch	2024	1831 t	1843 t
F	2024	0.26 (estimated F)	0.302 (assumed F)
Target F for TAC (F_{MSY})	2025	0.386	0.361

***Note '2024' = Intermediate year in the previous assessment; '2025' = advice year in the previous assessment**

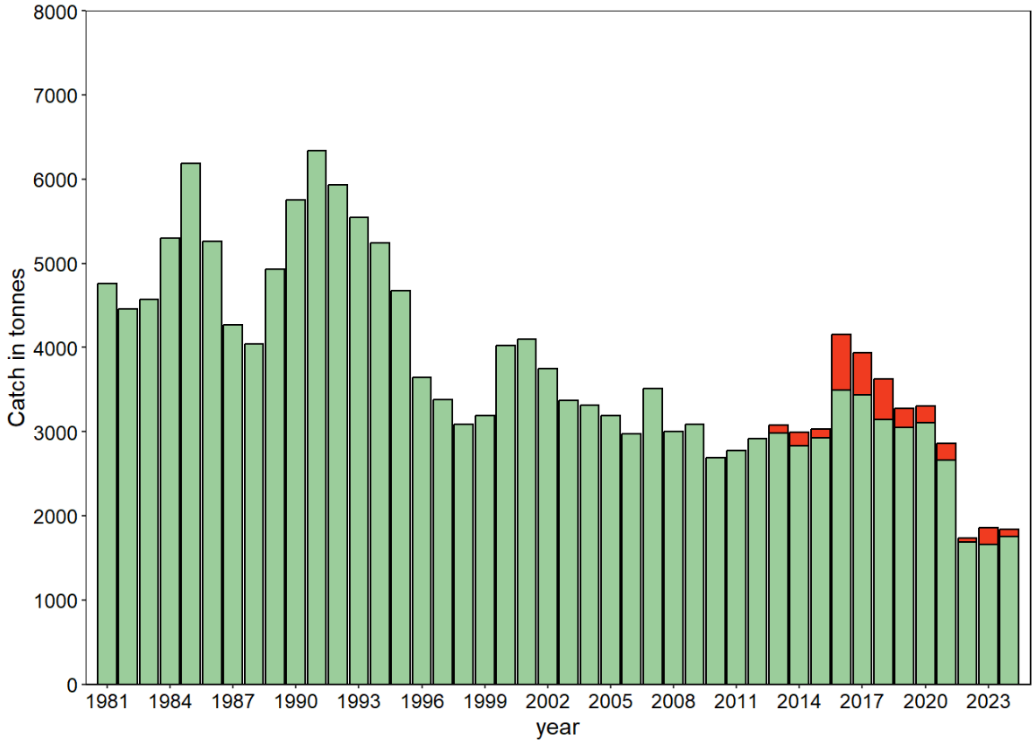

Figure 20.1. Turbot in Subarea 4. Total catches 1981–2024. ICES estimated landings (green) and discards (red).

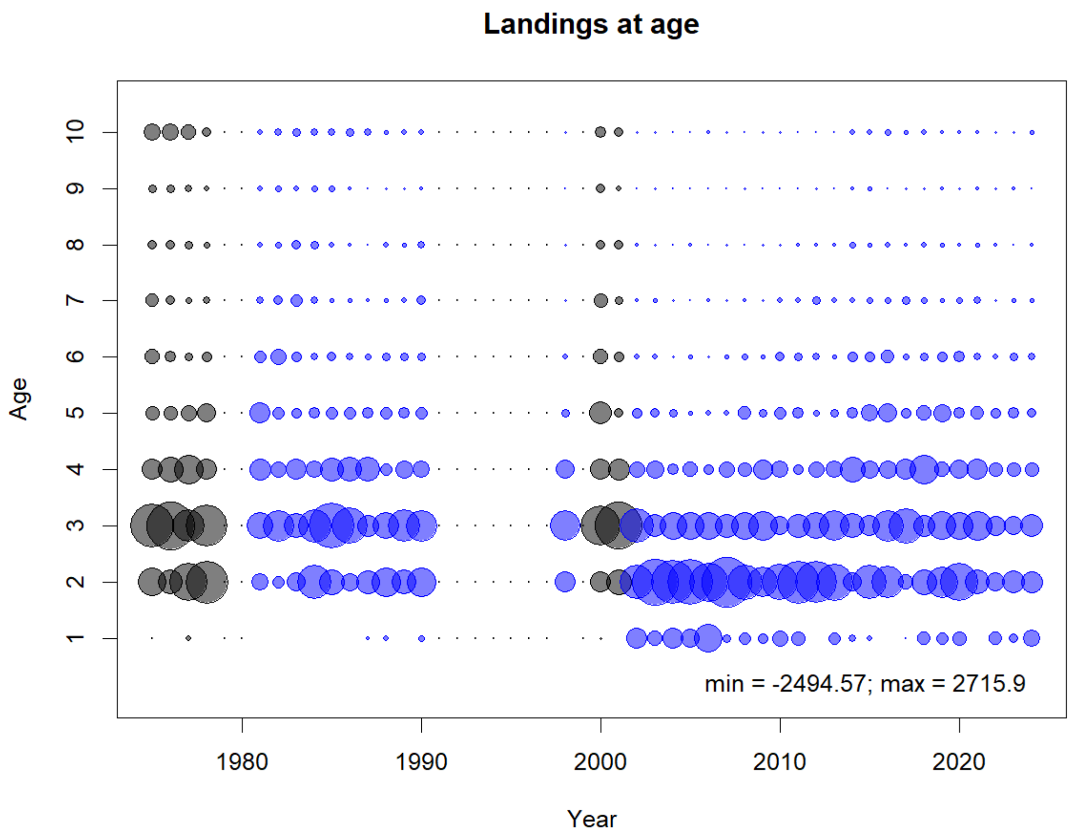

Figure 20.2. Turbot in Subarea 4. Landings-at-age for the years with available data between 1975–2024. Data for 1991–1997 and 1999–2002 are missing. Grey points indicate data not used in the assessment.

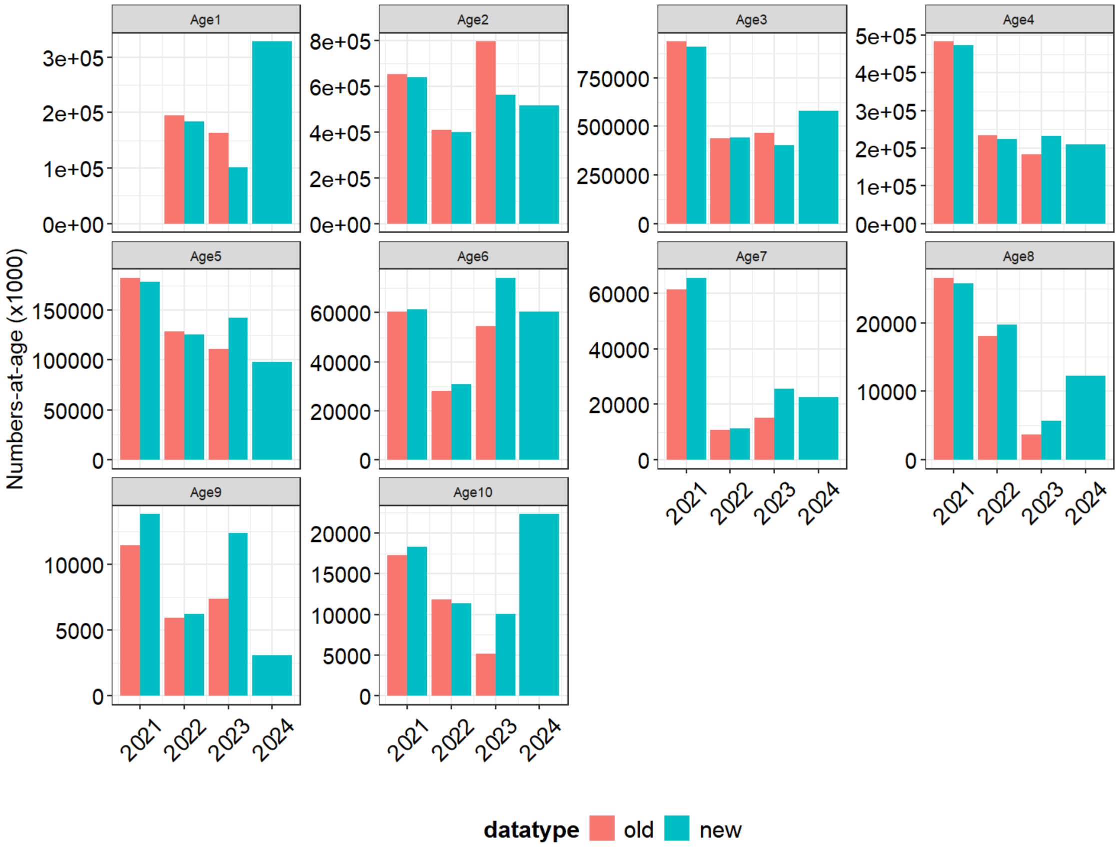

Figure 20.3. Turbot in Subarea 4. Re-calculated numbers at age for years 2021-2023 as well as the current age allocation distribution calculated for the 2024 data year used in WGNSSK 2024.

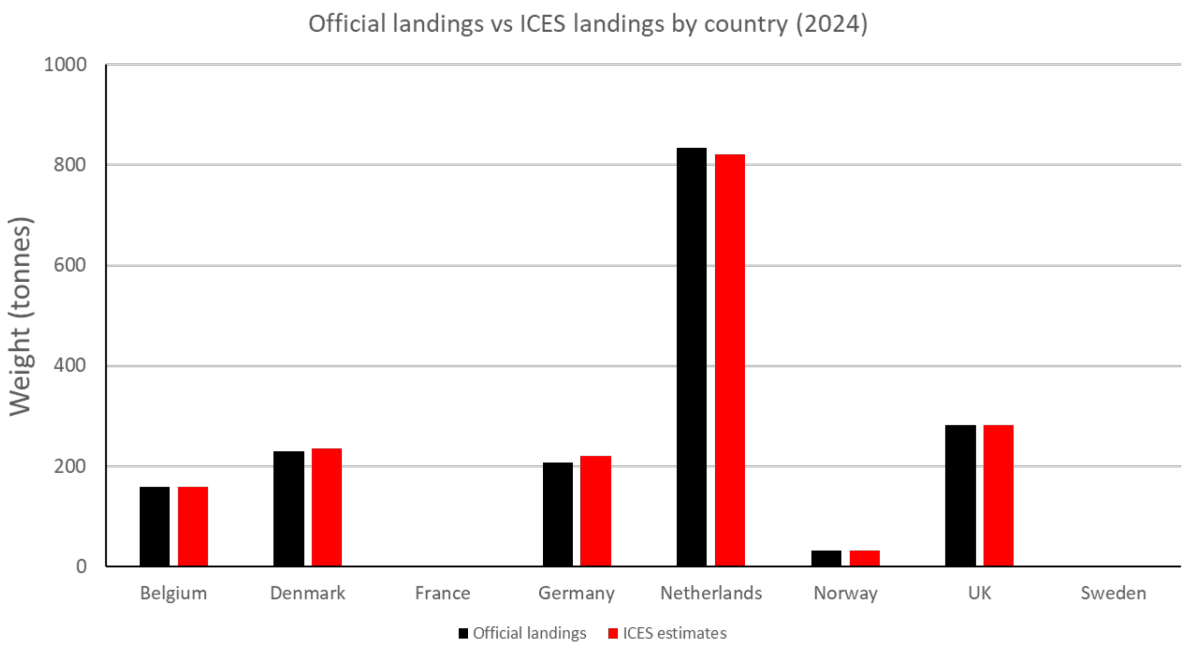

Figure 20.4. Turbot in Subarea 4. Comparison between official landings and ICES estimated landings between countries in the current assessment.

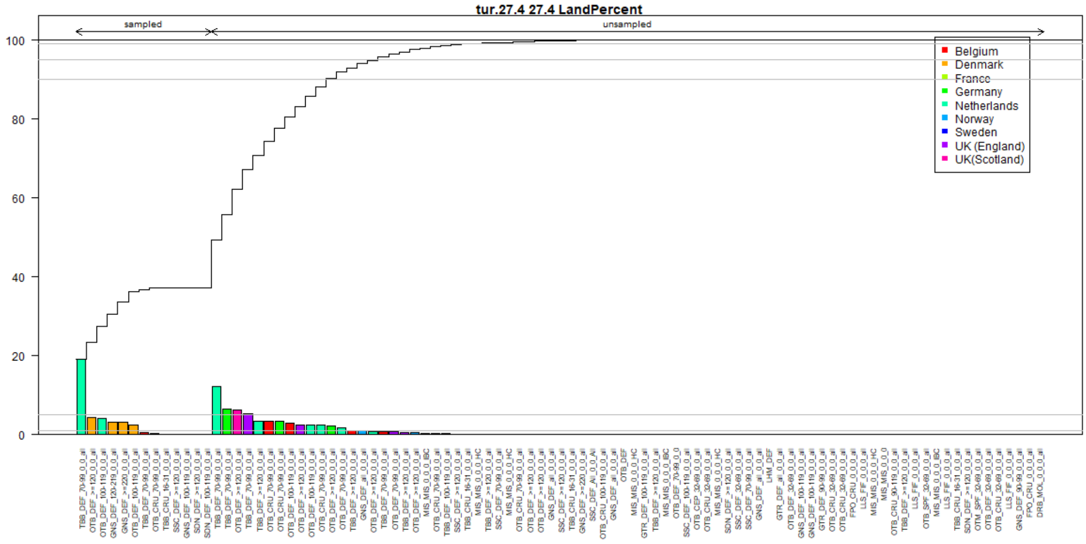

Figure 20.5. Turbot in Subarea 4. Total landings by métier in 2024 sorted by sampled/unsampled age distributions in InterCatch.

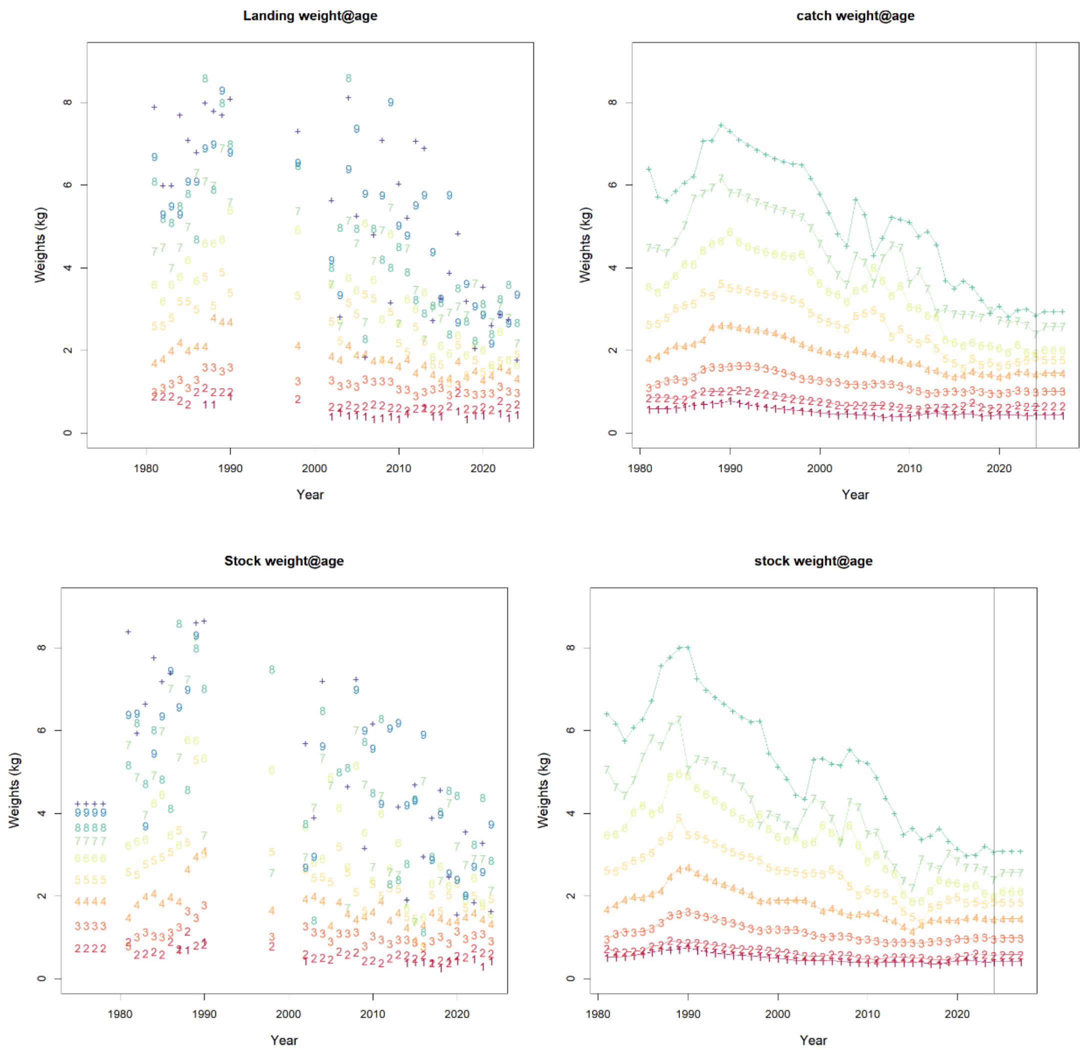

Figure 20.6. Turbot in Subarea 4. Raw landings (top-left), modelled landings (top right) and raw stock (bottom left) and modelled (bottom right) weight at age. Vertical line indicates last year in the assessment, and values used in the forecast are shown in the modelled weights-at-age plots.

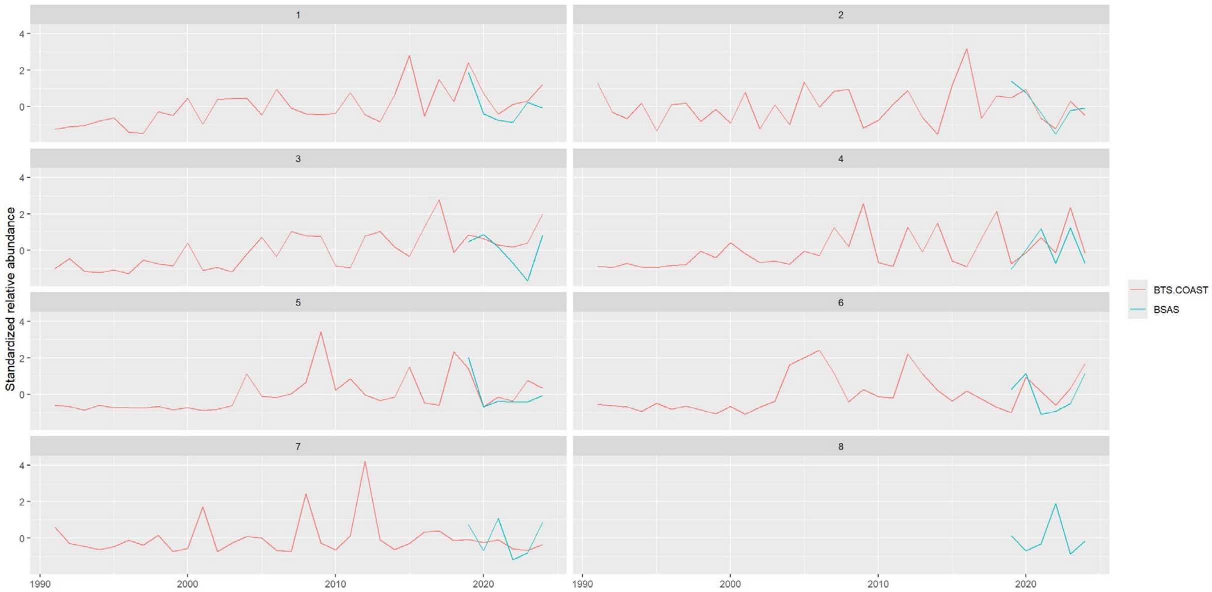

Figure 20.7. Turbot in Subarea 4. Time series of the standardized indices for ages 1 to 8 from the survey tuning fleets available for the assessment: BTS+COAST (1991-2024) and BSAS (2019-2024).

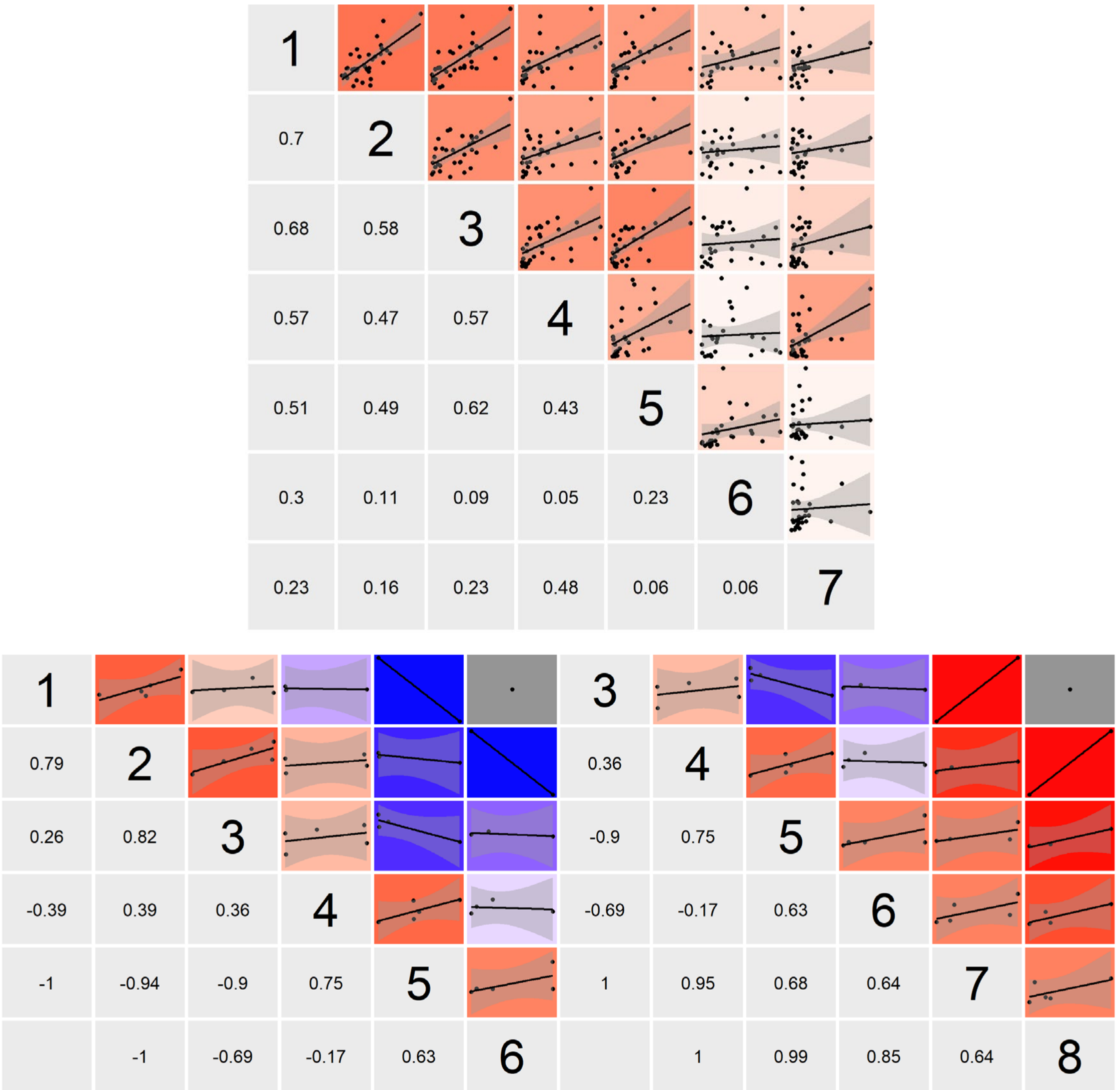

Figure 20.8. Turbot in Subarea 4. Internal consistency of the two tuning indices available for the assessment: BTS+COAST from 1991–2024 (top), and BSAS 2019–2024 (bottom).

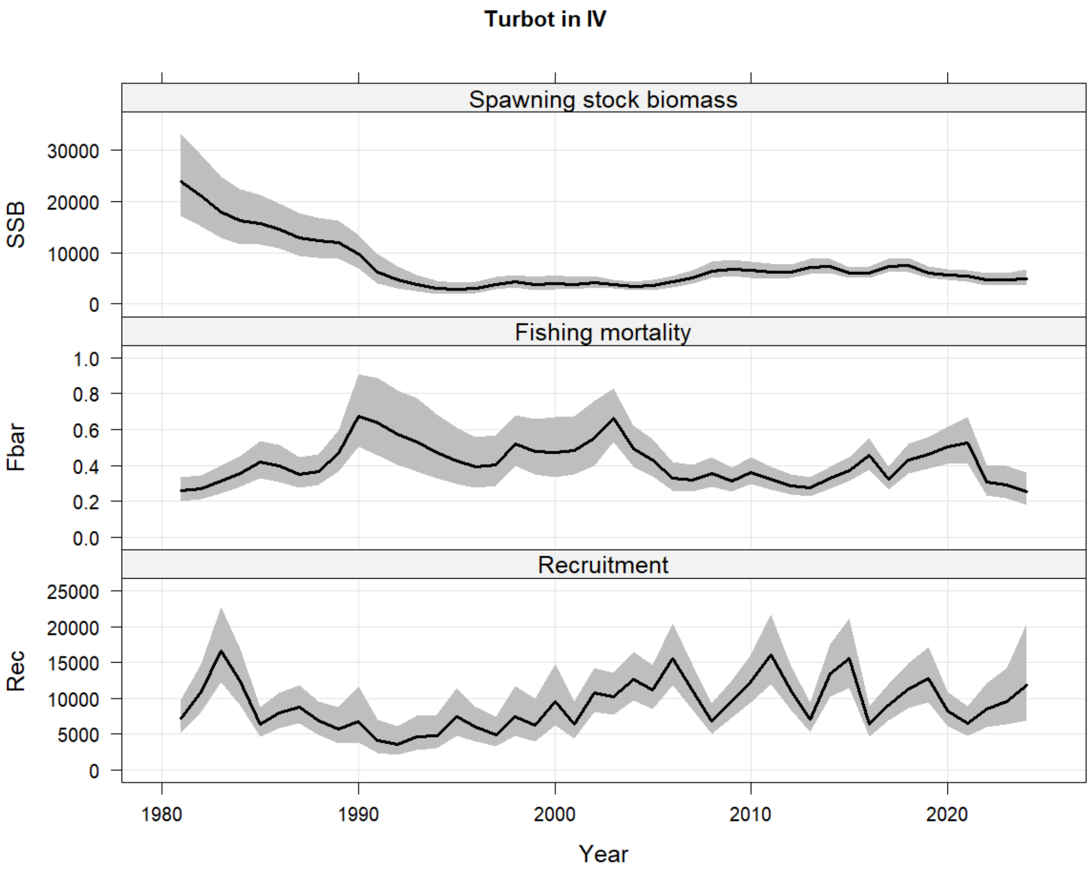

Figure 20.9. Turbot in Subarea 4. Summary plot of SSB, F and Recruitment, including the uncertainty bounds.

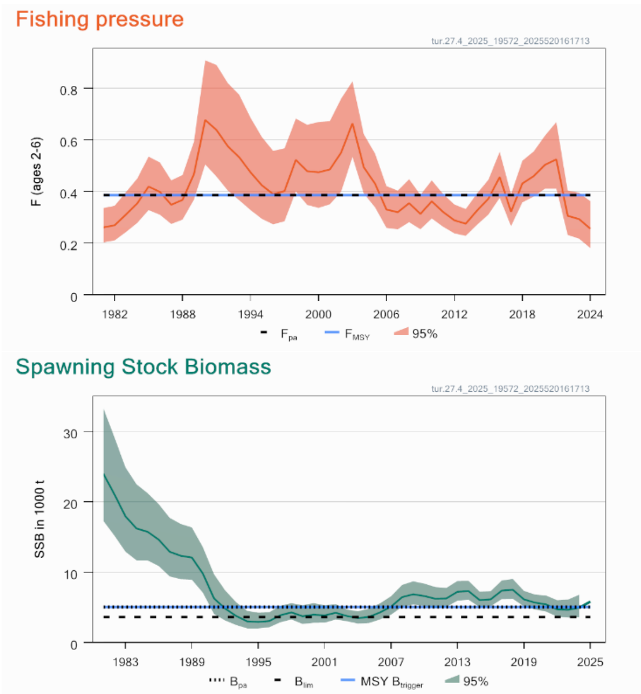

Figure 20.10. Turbot in Subarea 4. Summary of fishing pressure (top) and spawning stock biomass in relation to reference points.

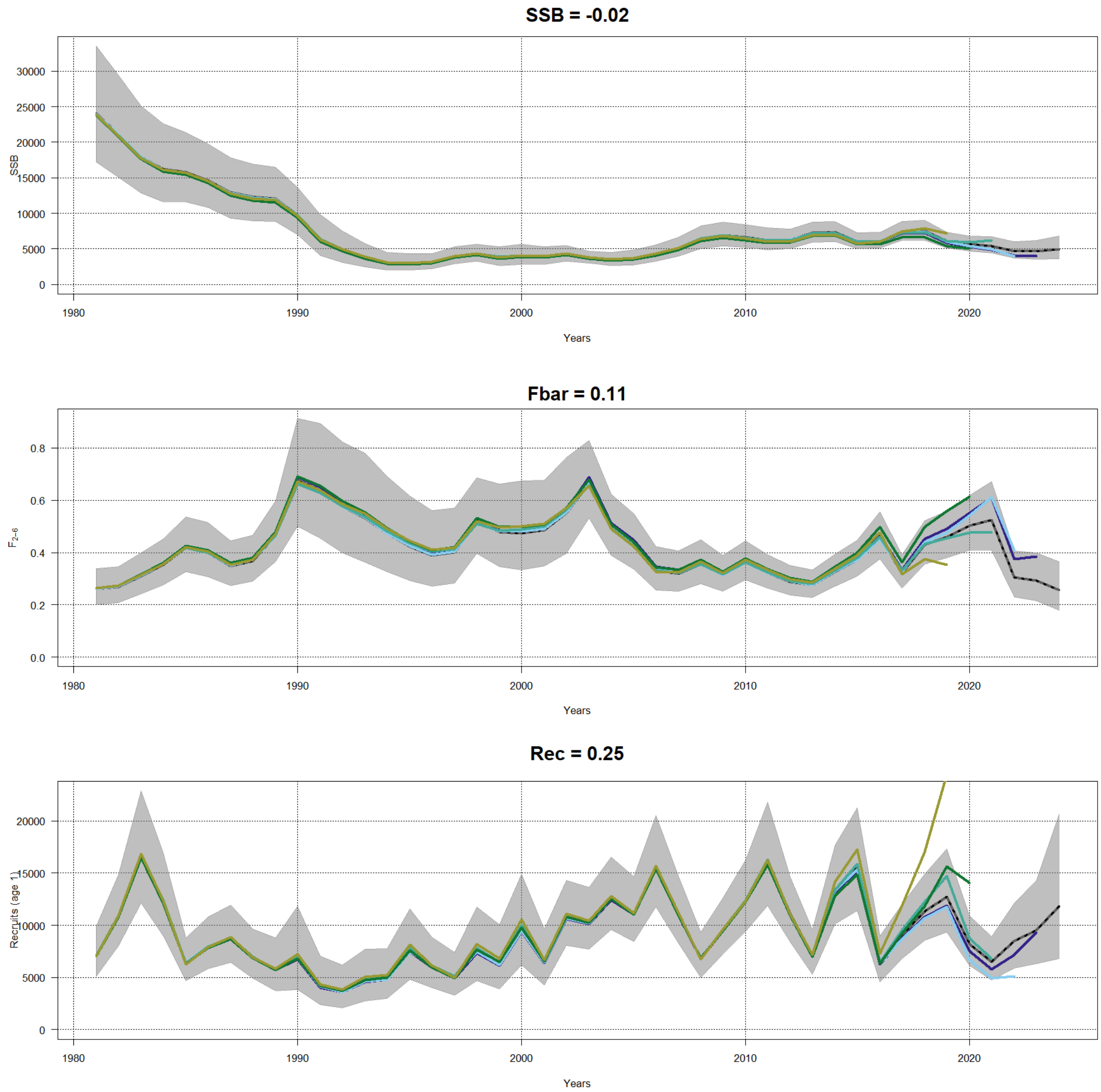

Figure 20.11. Turbot in Subarea 4. Retrospective analysis plot on SSB, F and R including the confidence band of the most recent assessment and Mohn's rho values.

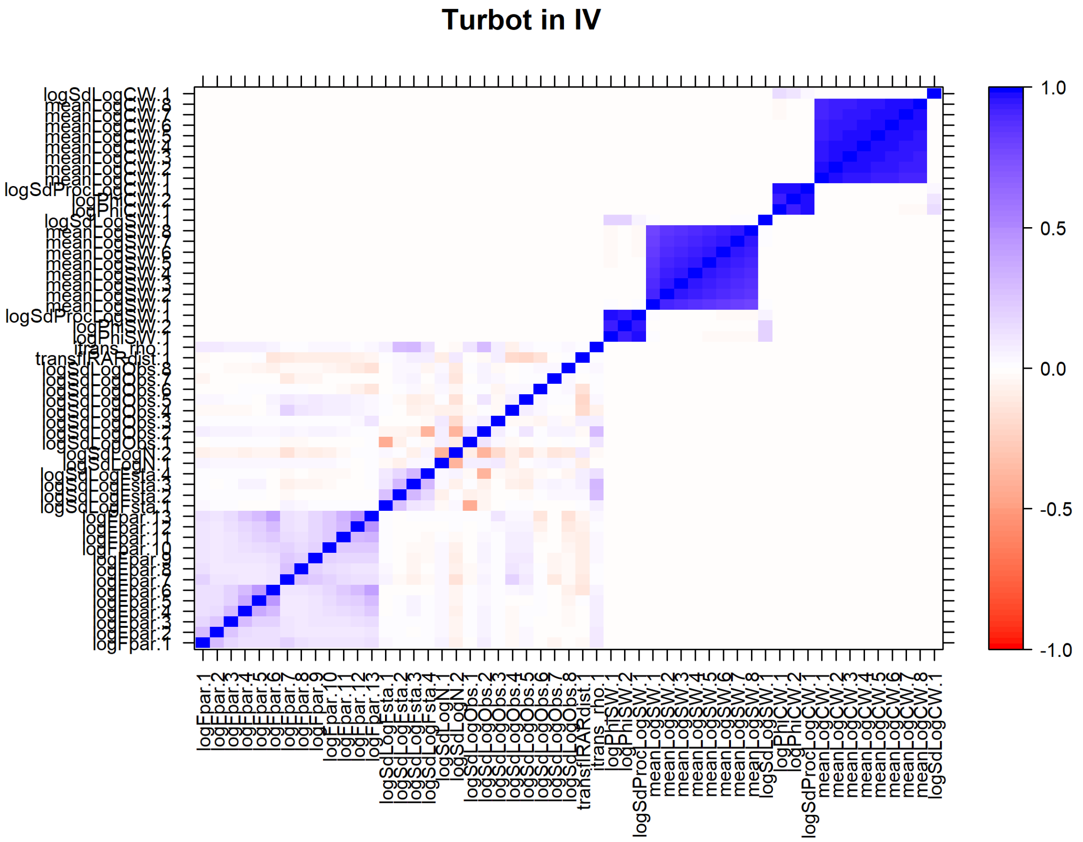

Figure 20.12. Turbot in Subarea 4. Parameter-correlation plot. It shows the correlation among all parameters that are estimated in the model. Fpar parameters refer to catchabilities, Fstates to the random walk in F, logN to the random walk in N, logObs to the observation variances, fRARdist to the auto-correlation in the surveys and trans_rho to the correlation in the F-random walks.

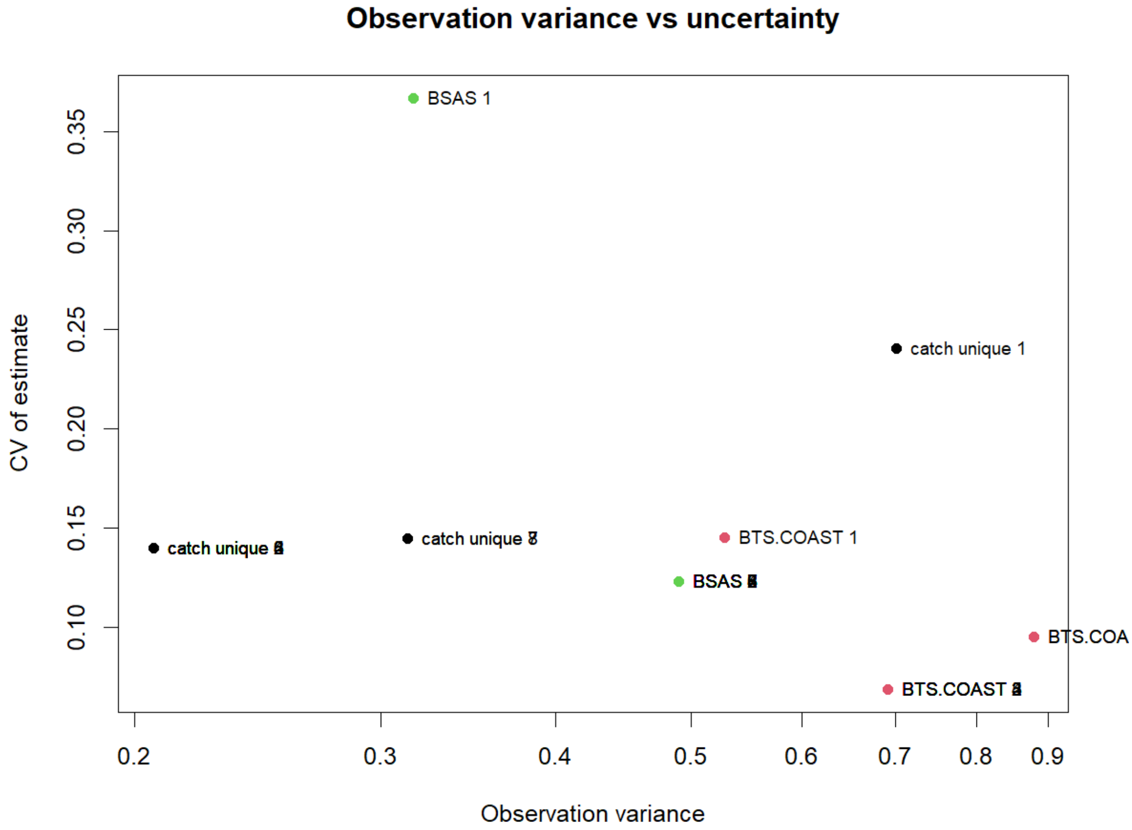

Figure 20.13. Turbot in Subarea 4. Plot showing the observation variance vs the CV of that estimate for the current assessment.

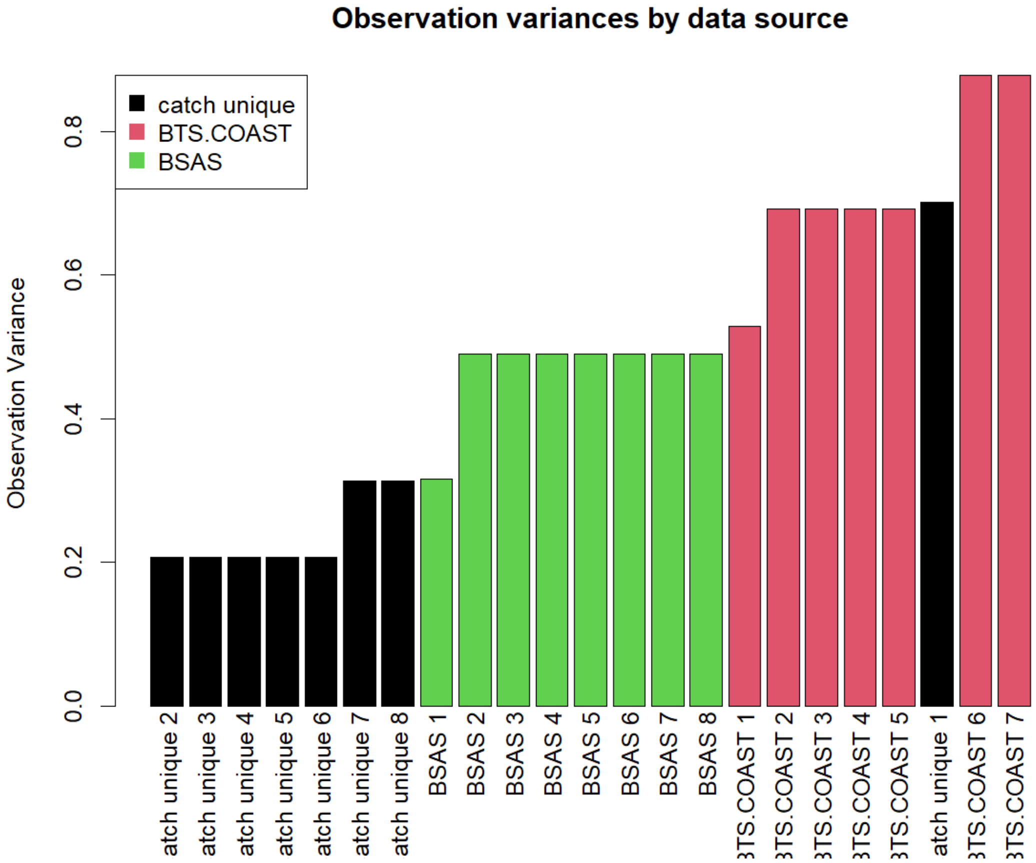

Figure 20.14. Turbot in Subarea 4. Estimated observation variances (scaling factor for each of the surveys), ordered from the best to the worst survey fit and has colour coding to show which bars belong to one dataset.

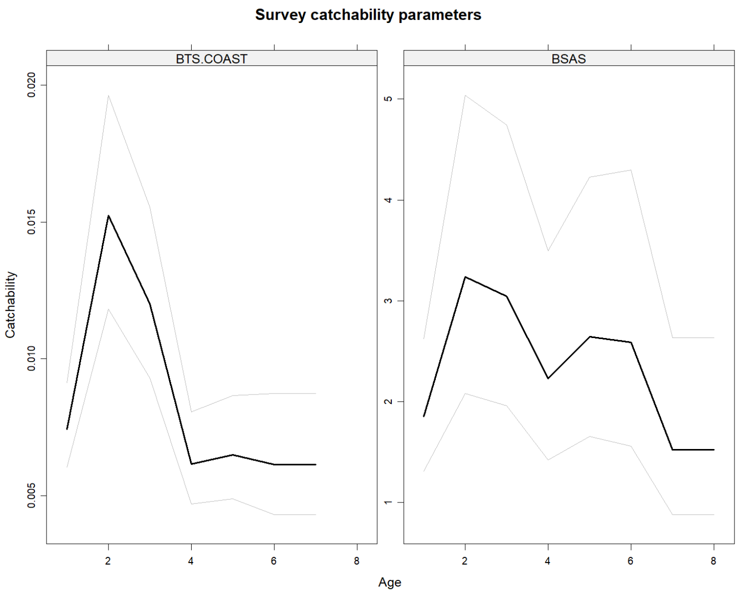

Figure 20.15. Turbot in Subarea 4. Catchabilities of the surveys for all surveys with more than 1 age-group.

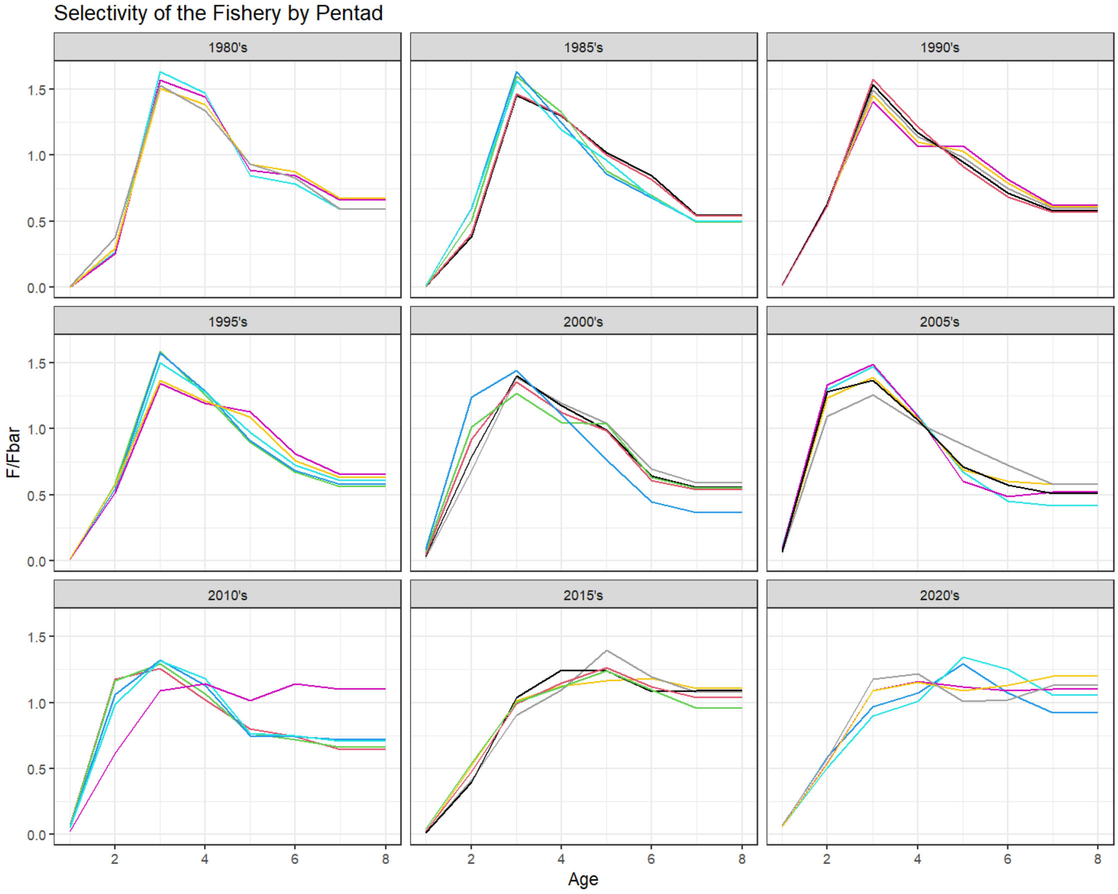

Figure 20.16. Turbot in Subarea 4. Estimated selectivity from 1981 to 2024, grouped by a 5-year period. Note the 1980s are 1981 up to 1984, 2015s is 2015 up to 2019. Values represent actual F-at-age.

Figure 20.17. Turbot in Subarea 4. Residual bubble plot of observation errors from the landings (top), BTS+COAST index (middle) and BSAS index (bottom).

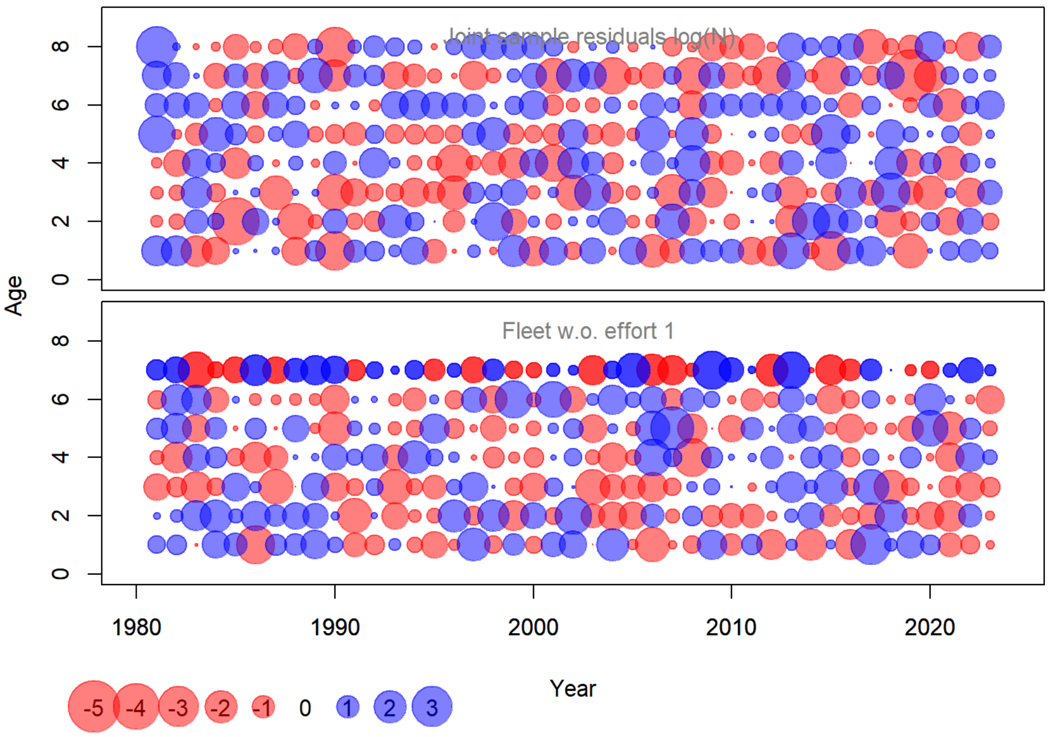

Figure 20.18. Turbot in Subarea 4. Residual bubble plot of process errors for log(N) and log(F) processes.

Figure 20.19. Turbot in Subarea 4. Leave-one-out analysis showing the effects from the two indices on spawning stock biomass (SSB), fishing mortality (Fbar) and recruitment. Could not be computed properly due to brevity of BSAS time series.

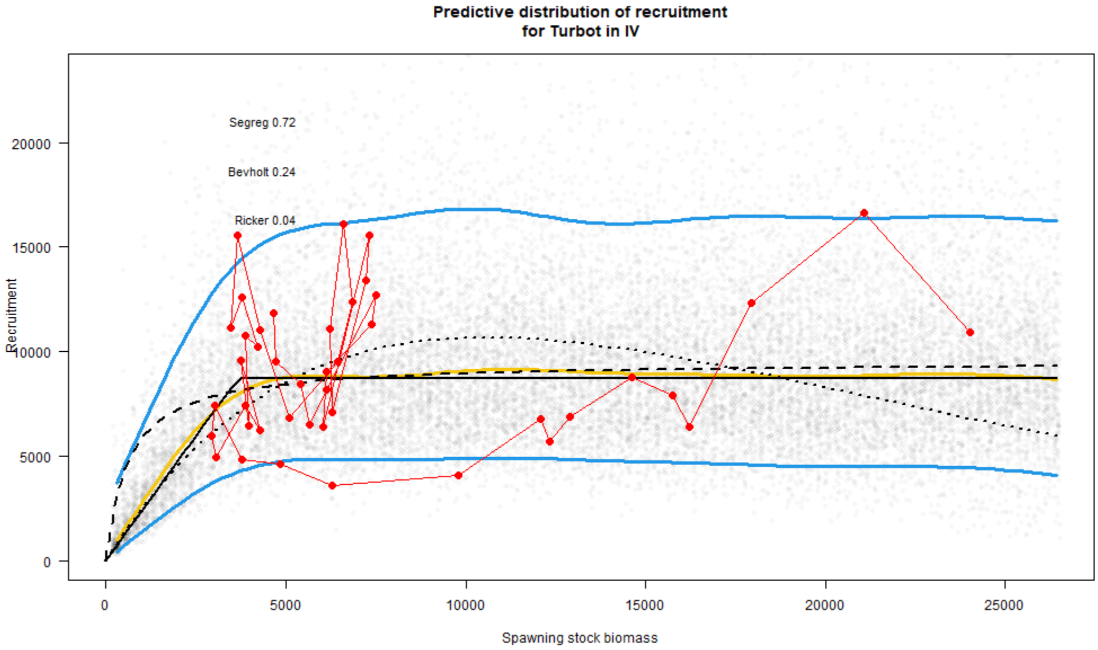

Figure 20.20. Turbot in Subarea 4. Stock recruitment pairs over time (1981-2024) which were used to calculate the reference points.

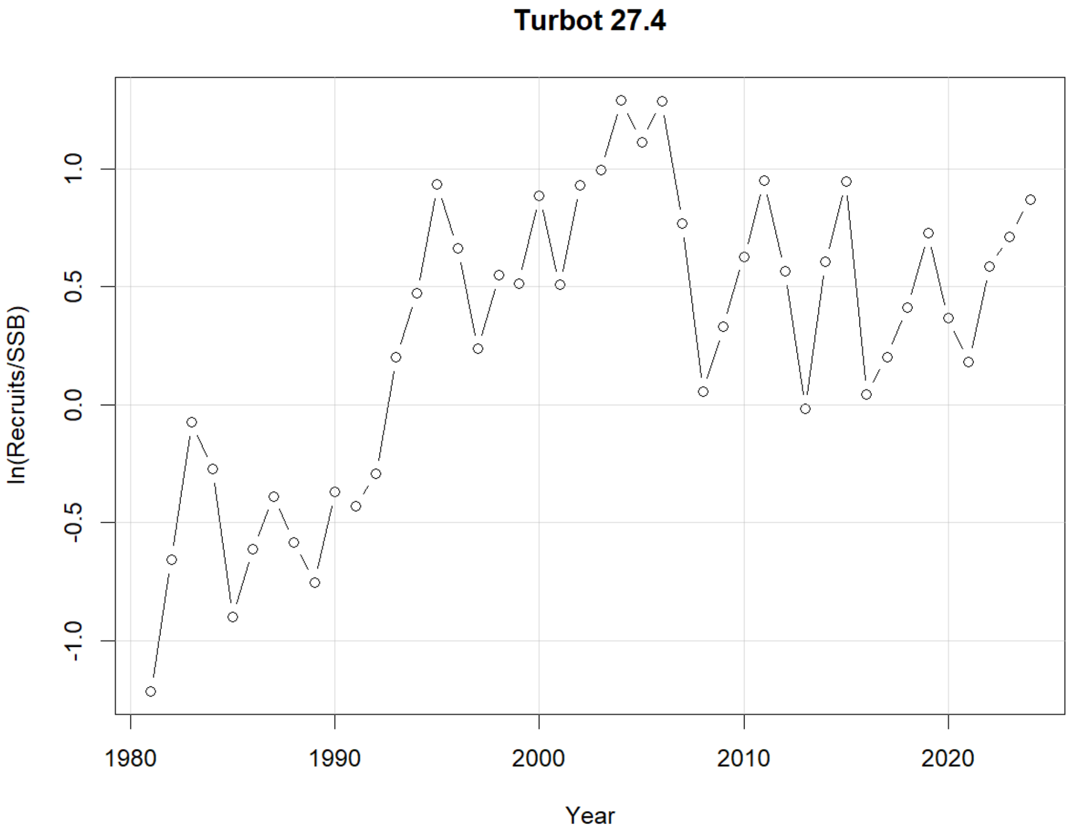

Figure 20.21. Turbot in Subarea 4. Productivity over time.

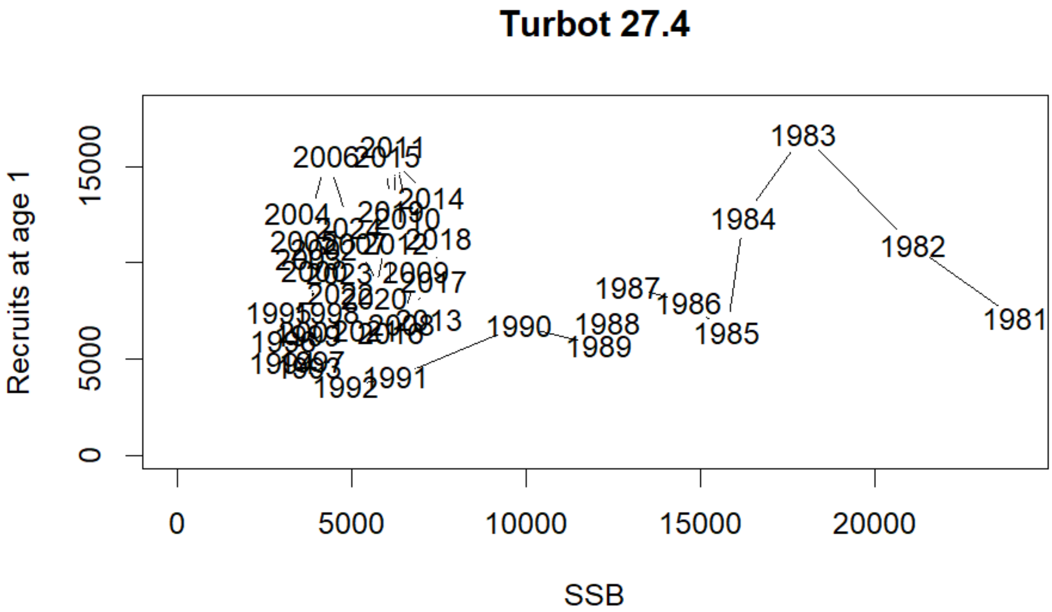

Figure 20.22. Turbot in Subarea 4. Stock recruitment pairs over time.

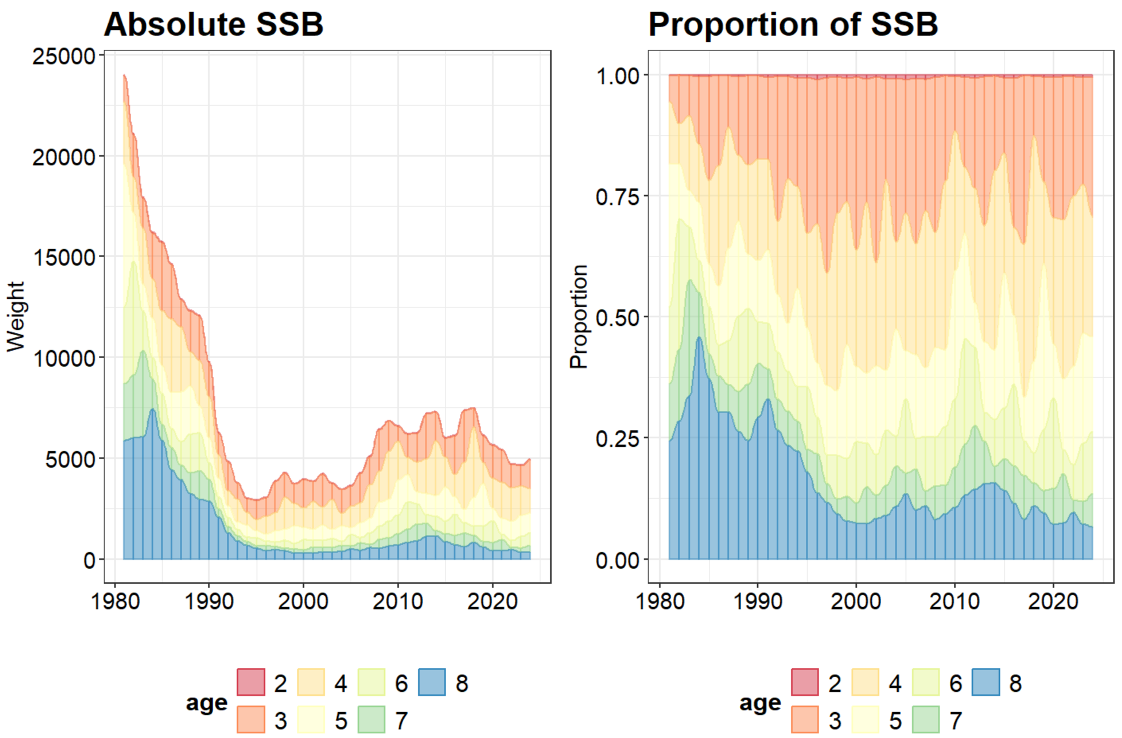

Figure 20.23. Turbot in Subarea 4. Model estimated absolute SSB-at-age (spawning stock biomass; *left*) and the estimated proportion of SSB-at-age (*right*).

Figure 20.24. Turbot in Subarea 4. The 3, 5, 10 and 20 year averages for selectivity-at-age in commercial catches (left) and modelled stock weights-at-age (*right*).

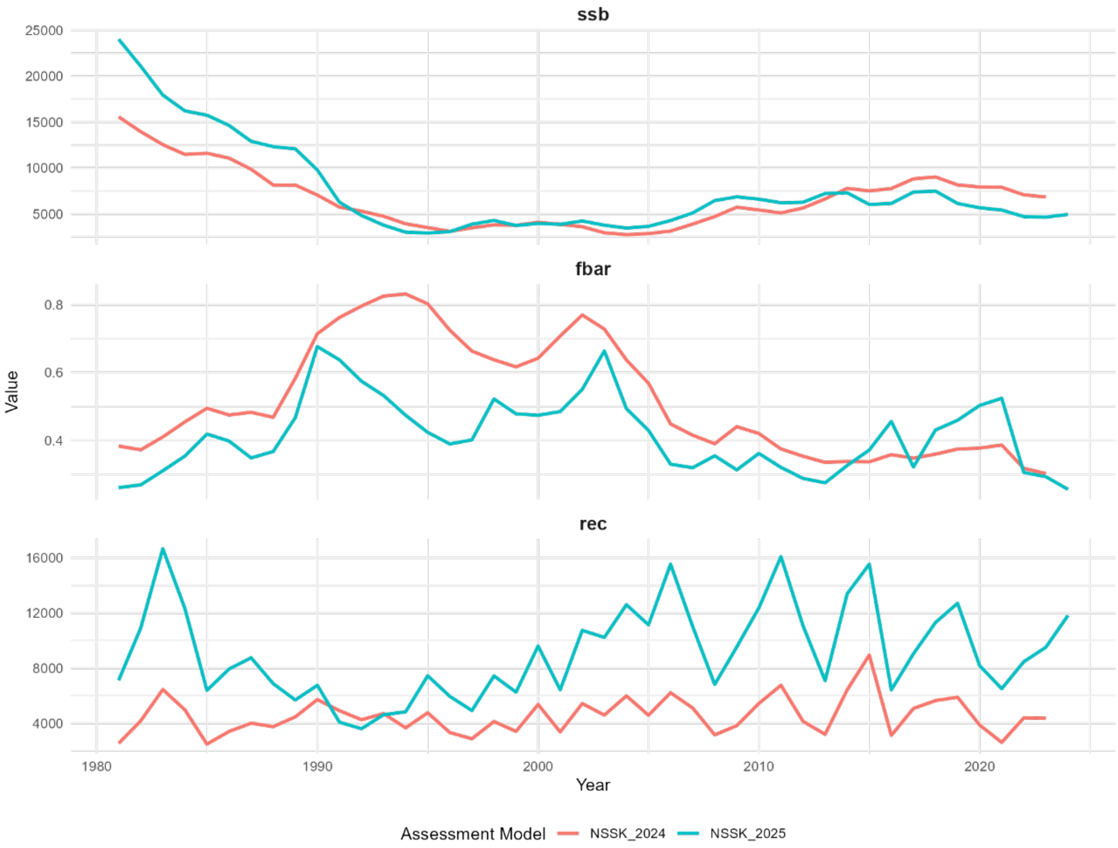

Figure 20.25. Turbot in Subarea 4. Differences between spawning stock biomass (ssb), fishing pressure (fbar), and recruitment (rec) between the previous years assessment and the current one. Note that this stock had a benchmark before the current assessment.

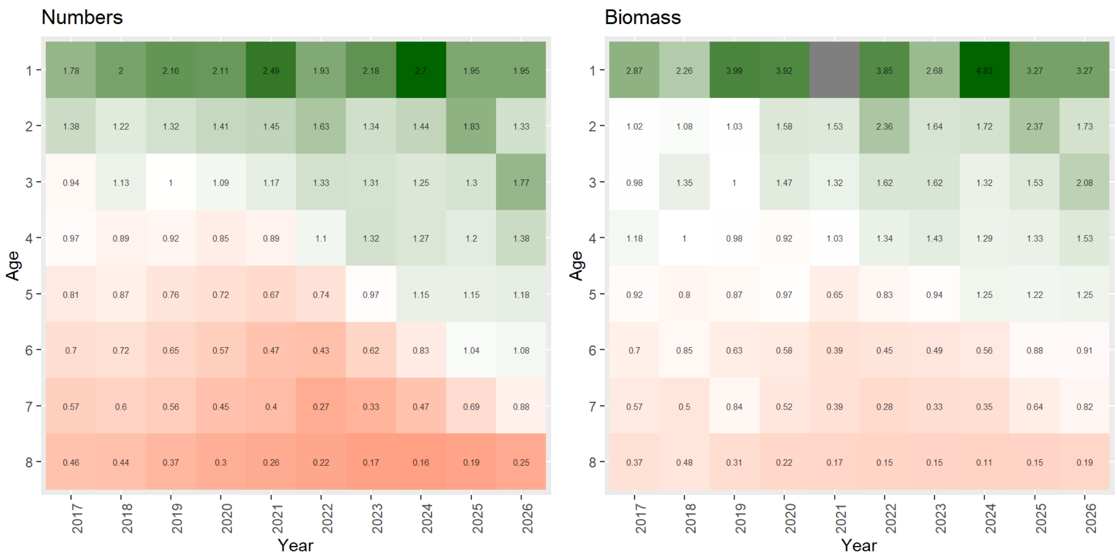

Figure 20.26. Turbot in Subarea 4. Ratios of numbers-at-age (left) and biomass-at-age (right) for WGNSSK 2025 to WGNSSK 2024 from years 2016 – 2025 (assessment and forecast). Note that this stock had a benchmark before the current assessment.

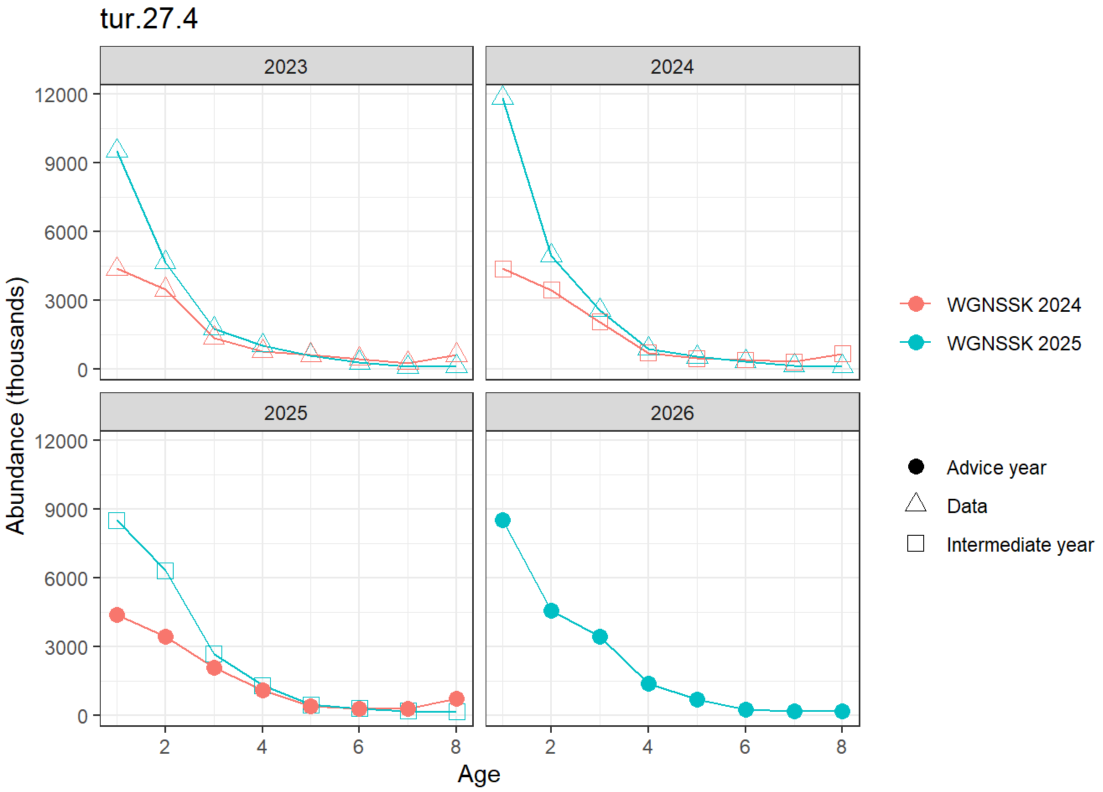

Figure 20.27. Turbot in Subarea 4. Comparison of abundances-at-age between WGNSSK 2024 and WGNSSK 2025 data and intermediate year forecasts for 2023-2025 as well as the 2026 advice year forecast from WGNSSK 2025. Note that this stock had a benchmark before the current assessment.

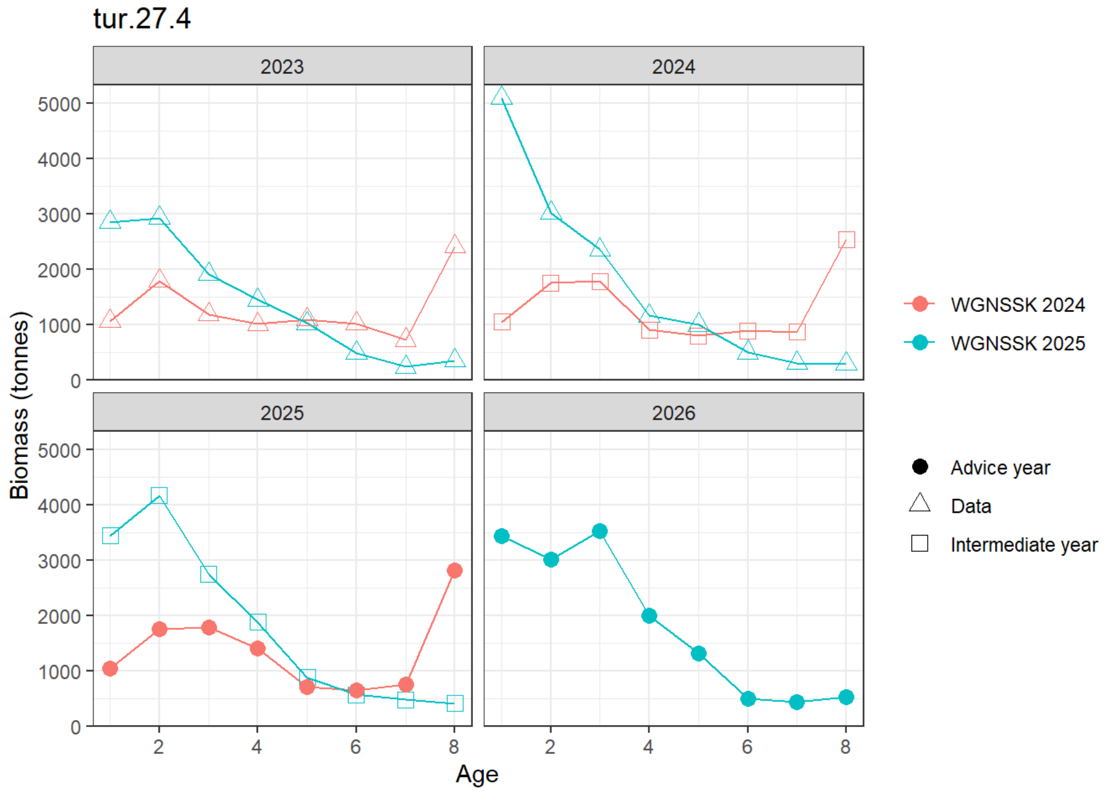

Figure 20.28. Turbot in Subarea 4. Comparison of biomass-at-age between WGNSSK 2024 and WGNSSK 2025 data and intermediate year forecasts for 2023-2025 as well as the 2026 advice year forecast from WGNSSK 2025. Note that this stock had a benchmark before the current assessment.

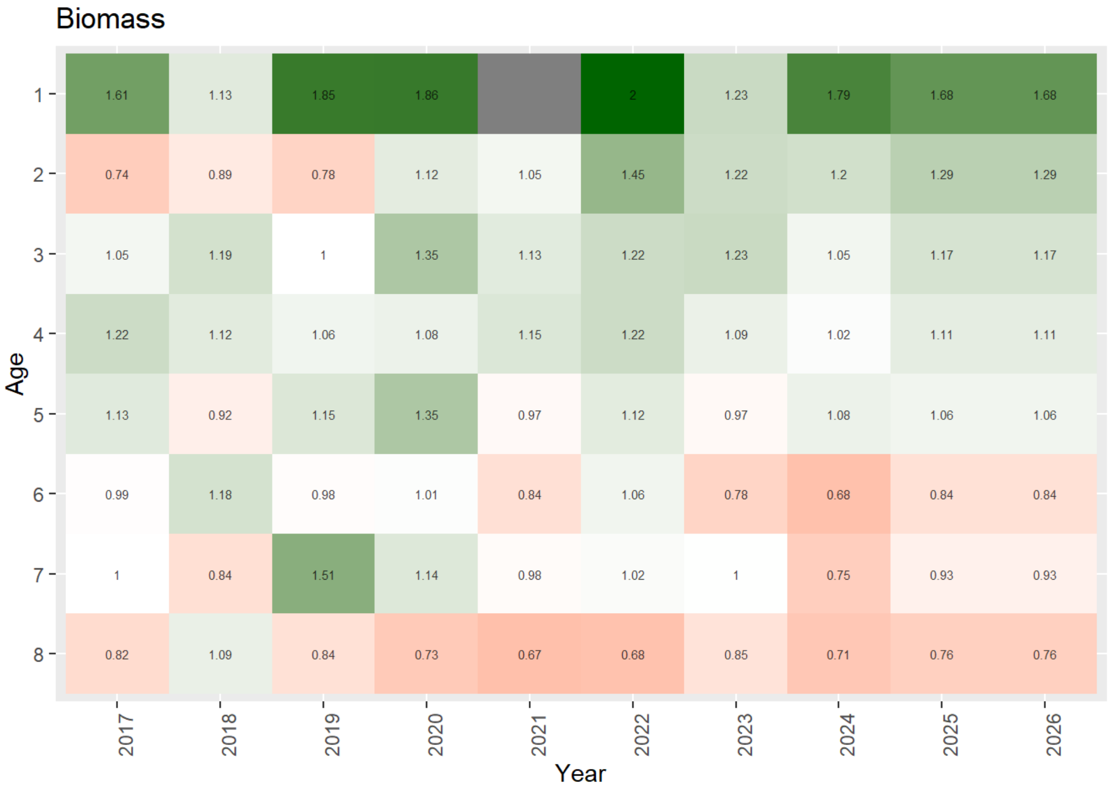

Figure 20.29. Turbot in Subarea 4. Ratios of weight-at-age for WGNSSK 2025 to WGNSSK 2024 from years 2017–2026 (assessment and forecast). Note that this stock had a benchmark before the current assessment.

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